Slow Controls Progress

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Outline of the Talk

- Design and test implementation progress
- Moving forward to final design and test: tasks and interfaces

Design and Test Progress

Slow Controls Design Scope

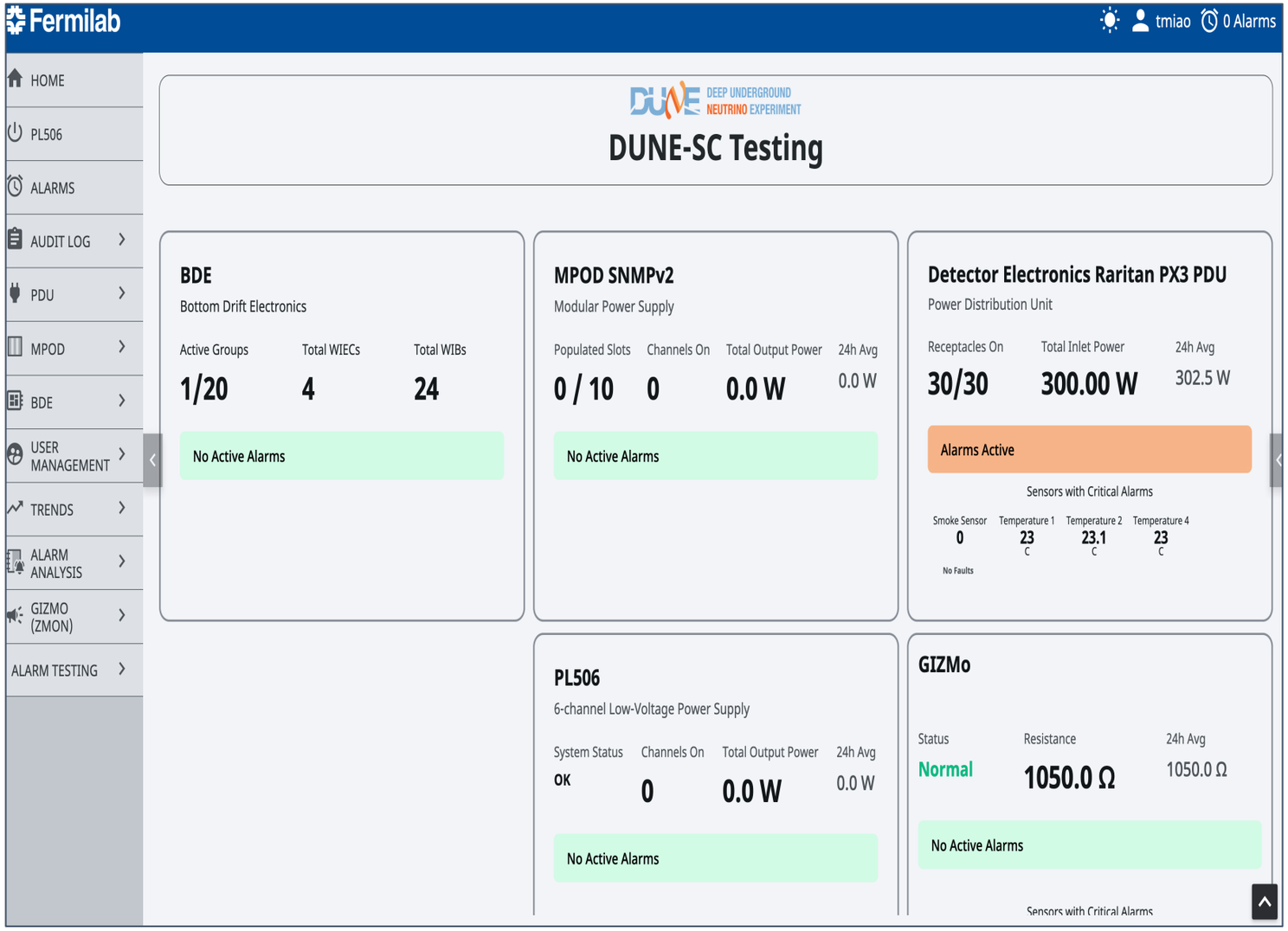
- Layout architecture
  - Control device list almost complete, except building equipment, environment sensors, RTDs in cryostat insulation
- Device connectivity
  - FD-CD has ~1700 control devices with native protocols SNMP, IMPI, ZMQ, LabVIEW, PLC… ➔ OPC-UA
  - FD-HD has similar number and types of native protocols
- Control and monitoring functions
  - GUI to display control functions and monitoring parameters specified by consortia in interfaces
  - Alarm generation and response functions
  - Automated control process for large group of devices (power on/off WIB/PTC/PL506, ISEGs…)
  - Detector protection interlocks: input to DPS as interlock conditions, inhibit from DPS to control devices
- Messaging and communications
  - Instant status report to and instruction from DAQ CCM, fast data to DQM

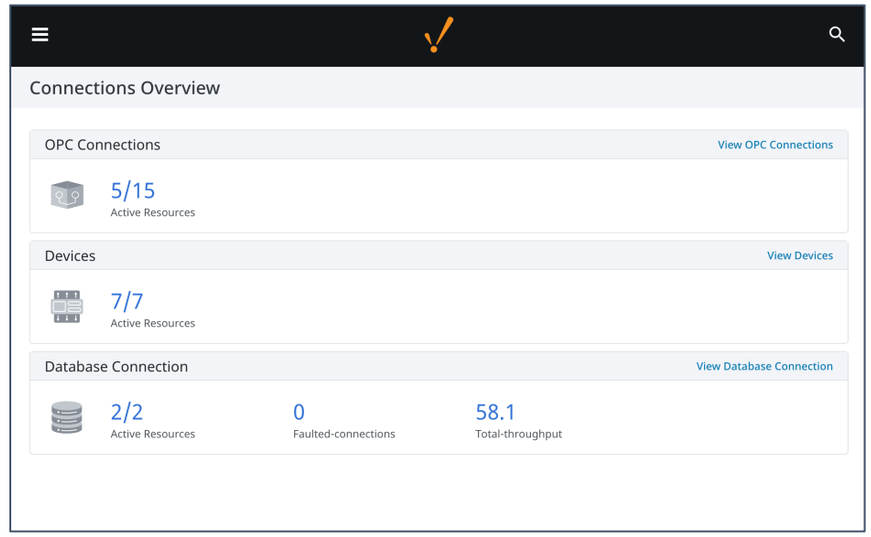
Design and Test Implementation: monitoring, control and automation

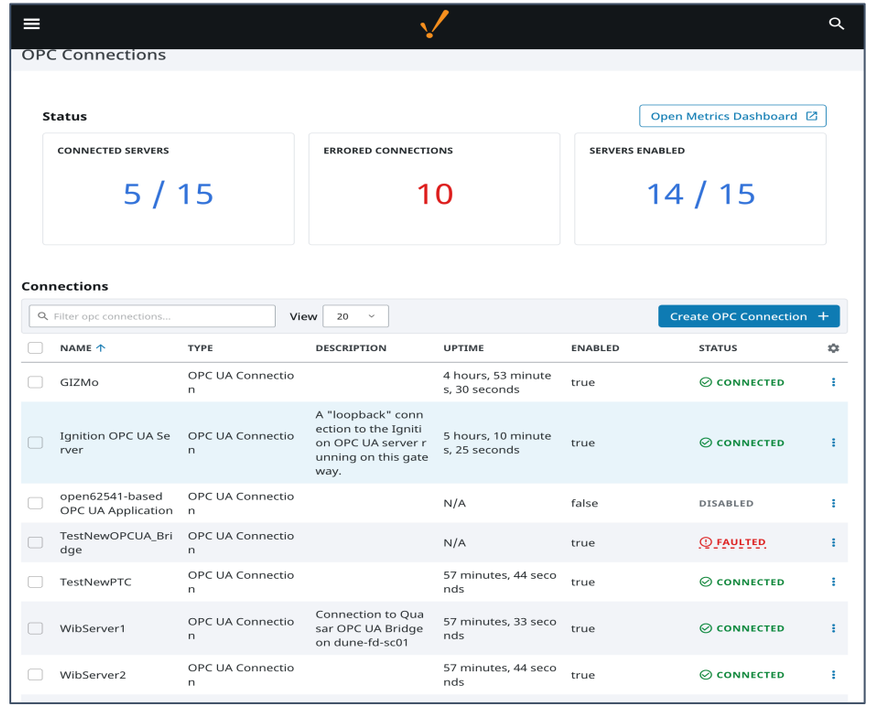
- Ignition provides tools for monitoring and control including alarm functions
  - HMIs and functions for displaying and manipulating data (trending, average, min/max…)
  - Built-in functions for alarm setting, response and history tracking
- Monitoring and control GUI design for subsystems
  - Slow controls is developing template using BDE control devices and a vertical slice test-stand at IERC
  - Consortia to use template to develop GUI for their specific needs ← *consortia participation is expected and critical*
- For large group devices (480 WIB, 80 PTC, 72/576 modules/channels ISEGs), we use SFC function to automate the power on/off process
  - Sequence Function Chart (SFC) is a visualized version of finite state machine process
  - Each transition between states checks device conditions using stored monitoring parameters (tags)
  - On-going SFC programming for PL506→PTC→WIB

DCS Project Page (screenshots)

https://dune-fd-sc01.fnal.gov:8043/data/perspective/client/Dune-FD-SC01/Views/Home/Home

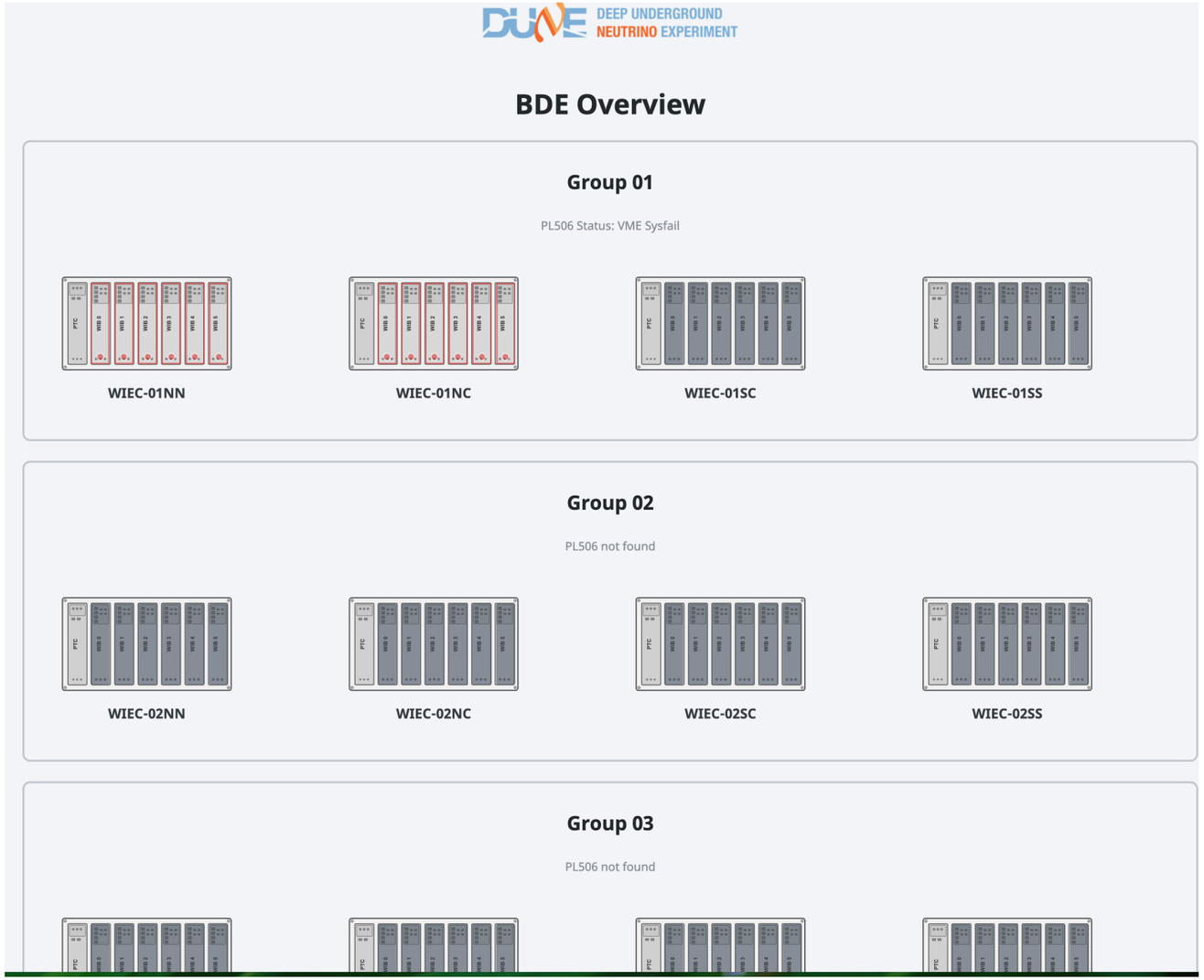






We will discuss alarm and response in later slides

BDE Project page (start page with WIEC crates)



GUI is organized according CRP naming scheme:
20 rows each has 4 WIECs and 1 PL506 for 4 CRPs

WIEC-01NN	WIEC-01NC	WIEC-01SC	WIEC-01SS	PL506-01	Power on/off CRP Row 1	DPS PLC I/O-01
WIEC-02NN	WIEC-02NC	WIEC-02SC	WIEC-02SS	PL506-02	Power on/off CRP Row 2	
WIEC-03NN	WIEC-03NC	WIEC-03SC	WIEC-03SS	PL506-03	Power on/off CRP Row 3	DPS PLC I/O-02
WIEC-04NN	WIEC-04NC	WIEC-04SC	WIEC-04SS	PL506-04	Power on/off CRP Row 4	
WIEC-05NN	WIEC-05NC	WIEC-05SC	WIEC-05SS	PL506-05	Power on/off CRP Row 5	DPS PLC I/O-03
WIEC-06NN	WIEC-06NC	WIEC-06SC	WIEC-06SS	PL506-06	Power on/off CRP Row 6	
WIEC-07NN	WIEC-07NC	WIEC-07SC	WIEC-07SS	PL506-07	Power on/off CRP Row 7	DPS PLC I/O-04
WIEC-08NN	WIEC-08NC	WIEC-08SC	WIEC-08SS	PL506-08	Power on/off CRP Row 8	
WIEC-09NN	WIEC-09NC	WIEC-09SC	WIEC-09SS	PL506-09	Power on/off CRP Row 9	DPS PLC I/O-05
WIEC-10NN	WIEC-10NC	WIEC-10SC	WIEC-10SS	PL506-10	Power on/off CRP Row 10	
WIEC-11NN	WIEC-11NC	WIEC-11SC	WIEC-11SS	PL506-11	Power on/off CRP Row 11	DPS PLC I/O-06
WIEC-12NN	WIEC-12NC	WIEC-12SC	WIEC-12SS	PL506-12	Power on/off CRP Row 12	
WIEC-13NN	WIEC-13NC	WIEC-13SC	WIEC-13SS	PL506-13	Power on/off CRP Row 13	DPS PLC I/O-07
WIEC-14NN	WIEC-14NC	WIEC-14SC	WIEC-14SS	PL506-14	Power on/off CRP Row 14	
WIEC-15NN	WIEC-15NC	WIEC-15SC	WIEC-15SS	PL506-15	Power on/off CRP Row 15	DPS PLC I/O-08
WIEC-16NN	WIEC-16NC	WIEC-16SC	WIEC-16SS	PL506-16	Power on/off CRP Row 16	
WIEC-17NN	WIEC-17NC	WIEC-17SC	WIEC-17SS	PL506-17	Power on/off CRP Row 17	DPS PLC I/O-09
WIEC-18NN	WIEC-18NC	WIEC-18SC	WIEC-18SS	PL506-18	Power on/off CRP Row 18	
WIEC-19NN	WIEC-19NC	WIEC-19SC	WIEC-19SS	PL506-19	Power on/off CRP Row 19	DPS PLC I/O-10
WIEC-20NN	WIEC-20NC	WIEC-20SC	WIEC-20SS	PL506-20	Power on/off CRP Row 20	

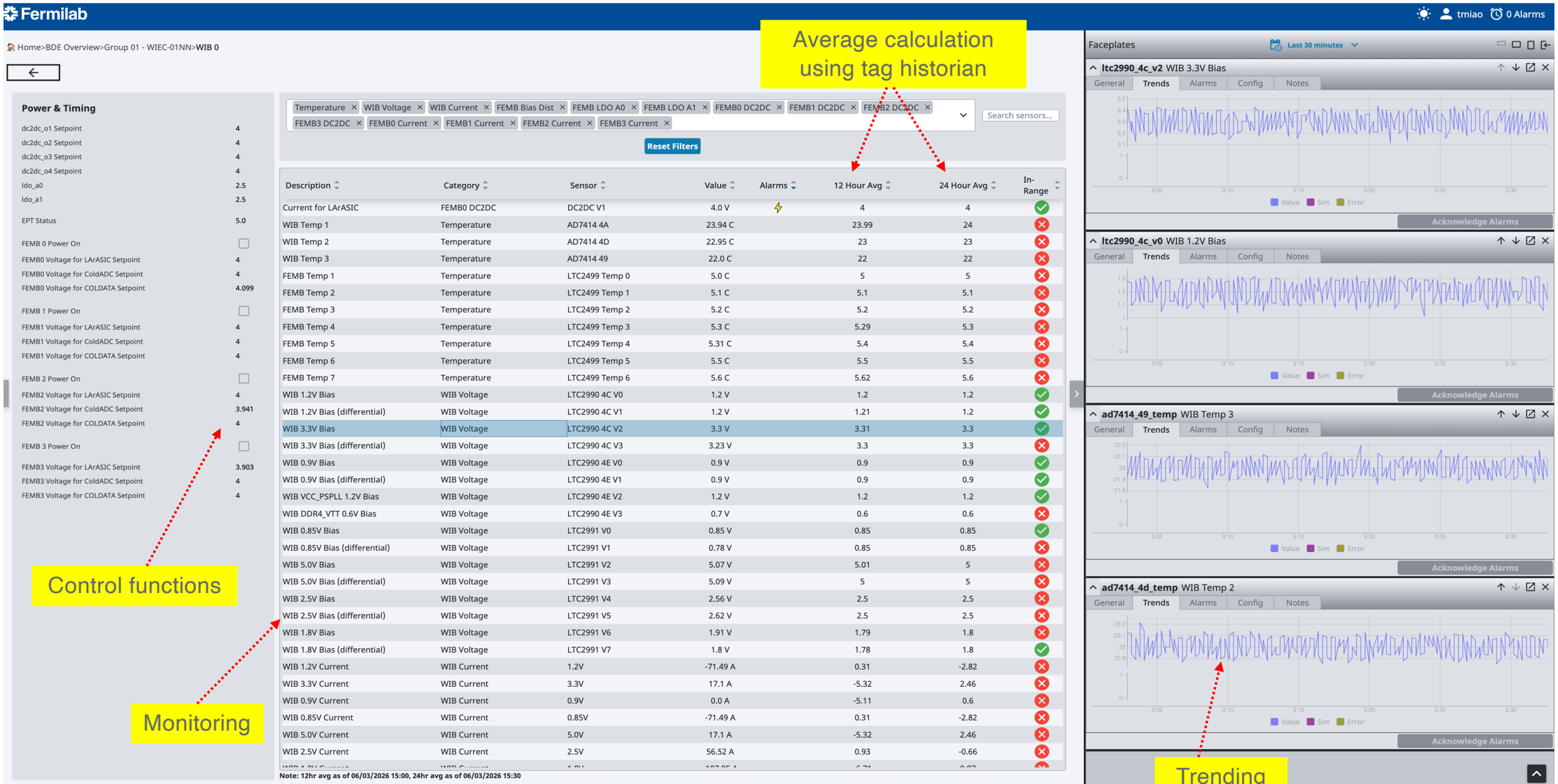
WIEC Page



Metric	WIB 0 ▶	WIB 1 ▶	WIB 2 ▶	WIB 3 ▶	WIB 4 ▶	WIB 5 ▶
FEMBs On	0/4	0/4	0/4	0/4	0/4	0/4
WIB Temp 1	24.0	24.0	24.0	24.0	24.0	24.0
WIB Temp 2	23.0	23.1	23.0	23.0	23.1	23.0
WIB Temp 3	22.1	22.0	22.0	21.9	22.0	22.0
FEMB Temp 1	5.1	4.9	5.0	5.0	5.0	5.0
FEMB Temp 2	5.1	5.1	5.2	5.1	5.1	5.1
FEMB Temp 3	5.2	5.1	5.1	5.2	5.2	5.2
FEMB Temp 4	5.4	5.4	5.3	5.3	5.3	5.2
FEMB Temp 5	5.4	5.4	5.4	5.5	5.4	5.5
FEMB Temp 6	5.6	5.5	5.5	5.5	5.6	5.6
FEMB Temp 7	5.6	5.6	5.5	5.7	5.6	5.5
1.2V Current	-57.77	0.0	0.0	0.0	0.0	0.0
3.3V Current	-24.13	0.0	0.0	0.0	0.0	0.0
0.9V Current	0.0	85.6	-114.09	72.08	0.0	0.0
0.85V Current	-57.77	0.0	0.0	0.0	0.0	0.0
5.0V Current	-24.13	0.0	0.0	0.0	0.0	0.0
2.5V Current	70.91	0.0	70.53	0.0	0.0	0.0
1.8V Current	0.0	0.0	52.01	0.0	1.28	0.0
Alarms (C/H/M/L)	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0	0/0/0/0

- Consortium defines what to show in this summary page
- PTC, PL506 to be added later

WIB Page


Fermilab
tmiao
0 Alarms
Home>BDE Overview>Group 01 - WIEC-01NN>WIB 0
Average calculation using tag historian
Power & Timing
dc2dc_o1 Setpoint 4
dc2dc_o2 Setpoint 4
dc2dc_o3 Setpoint 4
dc2dc_o4 Setpoint 4
ldo_a0 2.5
ldo_a1 2.5
EPT Status 5.0
FEMB 0 Power On
FEMB0 Voltage for LArASIC Setpoint 4
FEMB0 Voltage for ColdADC Setpoint 4
FEMB0 Voltage for COLDATA Setpoint 4.099
FEMB 1 Power On
FEMB1 Voltage for LArASIC Setpoint 4
FEMB1 Voltage for ColdADC Setpoint 4
FEMB1 Voltage for COLDATA Setpoint 4
FEMB 2 Power On
FEMB2 Voltage for LArASIC Setpoint 4
FEMB2 Voltage for ColdADC Setpoint 3.941
FEMB2 Voltage for COLDATA Setpoint 4
FEMB 3 Power On
FEMB3 Voltage for LArASIC Setpoint 3.903
FEMB3 Voltage for ColdADC Setpoint 4
FEMB3 Voltage for COLDATA Setpoint 4
Temperature
WIB Voltage
WIB Current
FEMB Bias Dist
FEMB LDO A0
FEMB LDO A1
FEMB0 DC2DC
FEMB1 DC2DC
FEMB2 DC2DC
FEMB3 DC2DC
FEMB0 Current
FEMB1 Current
FEMB2 Current
FEMB3 Current
Search sensors...
Reset Filters
Description
Category
Sensor
Value
Alarms
12 Hour Avg
24 Hour Avg
In-Range
Current for LArASIC FEMB0 DC2DC DC2DC V1 4.0 V 4 4
WIB Temp 1 Temperature AD7414 4A 23.94 C 23.99 24
WIB Temp 2 Temperature AD7414 4D 22.95 C 23 23
WIB Temp 3 Temperature AD7414 49 22.0 C 22 22
FEMB Temp 1 Temperature LTC2499 Temp 0 5.0 C 5 5
FEMB Temp 2 Temperature LTC2499 Temp 1 5.1 C 5.1 5.1
FEMB Temp 3 Temperature LTC2499 Temp 2 5.2 C 5.2 5.2
FEMB Temp 4 Temperature LTC2499 Temp 3 5.3 C 5.29 5.3
FEMB Temp 5 Temperature LTC2499 Temp 4 5.31 C 5.4 5.4
FEMB Temp 6 Temperature LTC2499 Temp 5 5.5 C 5.5 5.5
FEMB Temp 7 Temperature LTC2499 Temp 6 5.6 C 5.62 5.6
WIB 1.2V Bias WIB Voltage LTC2990 4C V0 1.2 V 1.2 1.2
WIB 1.2V Bias (differential) WIB Voltage LTC2990 4C V1 1.2 V 1.21 1.2
WIB 3.3V Bias WIB Voltage LTC2990 4C V2 3.3 V 3.31 3.3
WIB 3.3V Bias (differential) WIB Voltage LTC2990 4C V3 3.23 V 3.3 3.3
WIB 0.9V Bias WIB Voltage LTC2990 4E V0 0.9 V 0.9 0.9
WIB 0.9V Bias (differential) WIB Voltage LTC2990 4E V1 0.9 V 0.9 0.9
WIB VCC_PSPLL 1.2V Bias WIB Voltage LTC2990 4E V2 1.2 V 1.2 1.2
WIB DDR4_VTT 0.6V Bias WIB Voltage LTC2990 4E V3 0.7 V 0.6 0.6
WIB 0.85V Bias WIB Voltage LTC2991 V0 0.85 V 0.85 0.85
WIB 0.85V Bias (differential) WIB Voltage LTC2991 V1 0.78 V 0.85 0.85
WIB 5.0V Bias WIB Voltage LTC2991 V2 5.07 V 5.01 5
WIB 5.0V Bias (differential) WIB Voltage LTC2991 V3 5.09 V 5 5
WIB 2.5V Bias WIB Voltage LTC2991 V4 2.56 V 2.5 2.5
WIB 2.5V Bias (differential) WIB Voltage LTC2991 V5 2.62 V 2.5 2.5
WIB 1.8V Bias WIB Voltage LTC2991 V6 1.91 V 1.79 1.8
WIB 1.8V Bias (differential) WIB Voltage LTC2991 V7 1.8 V 1.78 1.8
WIB 1.2V Current WIB Current 1.2V -71.49 A 0.31 -2.82
WIB 3.3V Current WIB Current 3.3V 17.1 A -5.32 2.46
WIB 0.9V Current WIB Current 0.9V 0.0 A -5.11 0.6
WIB 0.85V Current WIB Current 0.85V -71.49 A 0.31 -2.82
WIB 5.0V Current WIB Current 5.0V 17.1 A -5.32 2.46
WIB 2.5V Current WIB Current 2.5V 56.52 A 0.93 -0.66
Note: 12hr avg as of 06/03/2026 15:00, 24hr avg as of 06/03/2026 15:30
Control functions
Monitoring
Faceplates
Last 30 minutes
ltc2990_4c_v2 WIB 3.3V Bias
ltc2990_4c_v0 WIB 1.2V Bias
ad7414_49_temp WIB Temp 3
ad7414_4d_temp WIB Temp 2
General
Trends
Alarms
Config
Notes
Value
Sim
Error
Acknowledge Alarms
Trending plots

Alarm Generation and Response

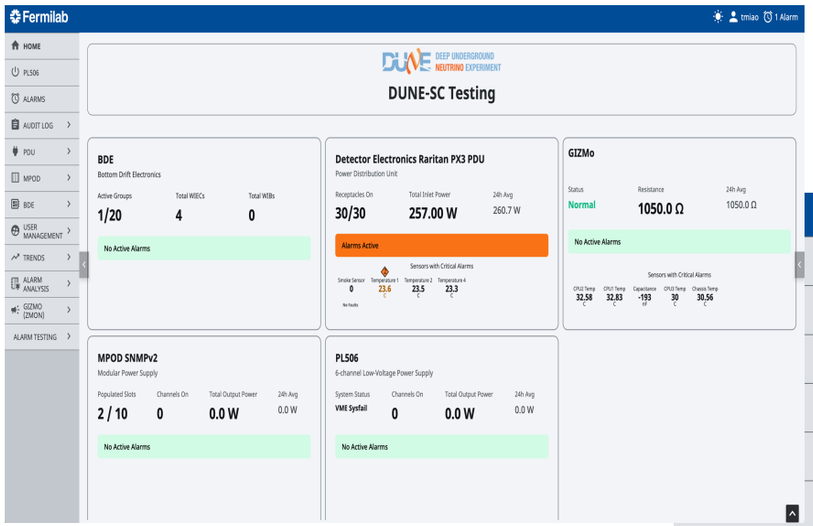
- Alarm generation
  - Selecting key parameters to generate alarms by consortium – see list for WIB at next page
  - Setting alarm threshold (mode): above/below; between; outside a window; not equal etc.
  - Classify alarm priority level: diagnostic, low, medium, high and critical
- Alarm response functions for expert to use for real-time activity
  - Acknowledgement: entered by experts indicating they are fixing underline issues
  - Shelving: expert to specify a time duration a particular alarm to be temporally excluded from alarm table
  - Custom functions for expert to modify alarm threshold, resetting priority level
- Alarm event log and history archiving
  - GUI for expert to write alarm handling log (in faceplate menu for example)
  - Ignition alarm journey → database as part of detector slow control status report

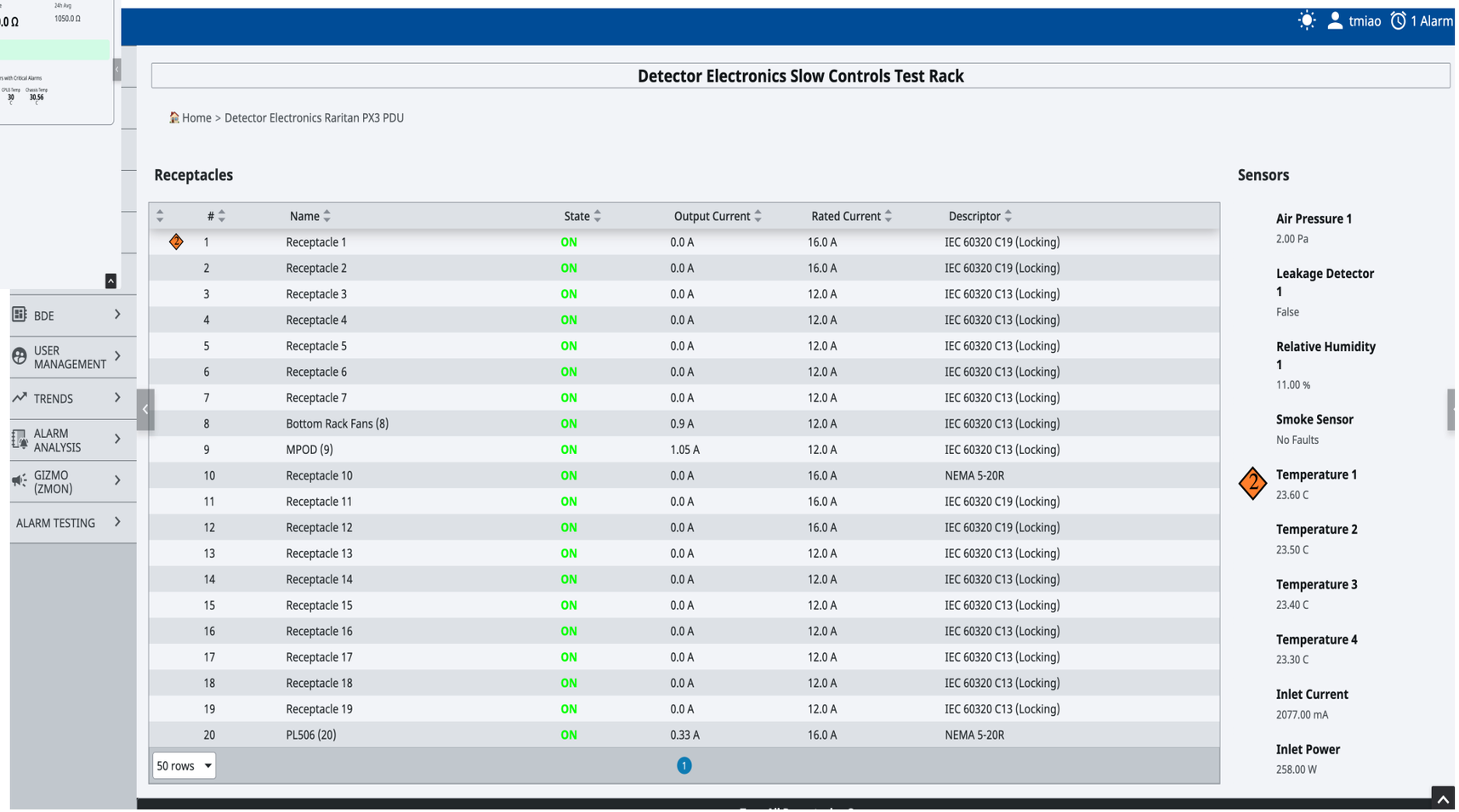
Key Parameters for Alarm Generation – example for WIB

Warm Interface Board and Frontend MotherBoard Control and Monitoring Parameters

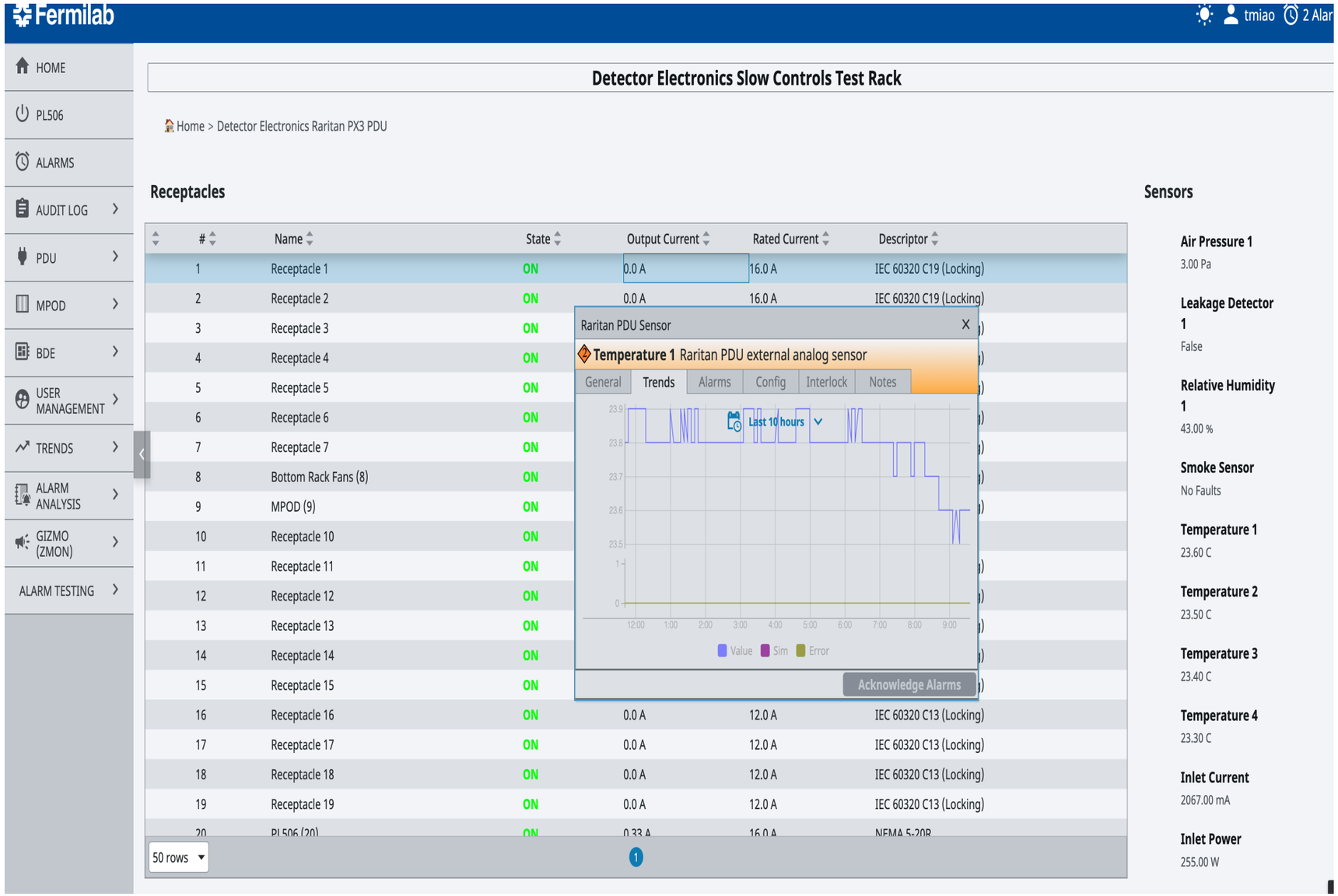
Hardware Device	Description	OPC-UA Address Name	Data Type	Nominal Value	Lower Bound	High Bound	Included in alarm table	Alarm Mode	Can be Monitored (Yes/No)	Configurable by DCS (Yes/No)
WIB	Firmware version	GetFWVersion	Uint32						Yes	No
WIB	WIB Status Bits	GetWIBStatus	Uint32						Yes	No
WIB	Temperature Sensors	Sensors	float	60 C	30 C	80 C	Yes	Above Setpoint	Yes	No
WIB	WIB 1.2 V Bias	Sensors	float	1.2 V	1.15 V	1.25 V			Yes	No
WIB	WIB 3.3 V Bias	Sensors	float	3.3 V	3.25 V	3.35 V			Yes	No
WIB	WIB 0.9 V Bias	Sensors	float	0.9 V	0.85 V	0.95 V			Yes	No
WIB	WIB VCC_PSPLL 1.2 V Bias	Sensors	float	1.2 V	1.15 V	1.25 V			Yes	No
WIB	WIB DDR4_VTT 0.6 V Bias	Sensors	float	0.6 V	0.55 V	0.65 V			Yes	No
WIB	WIB 0.85 V Bias	Sensors	float	0.85 V	0.80 V	0.90 V			Yes	No
WIB	WIB 5.0 V Bias	Sensors	float	5 V	4.95 V	5.05 V			Yes	No
WIB	WIB 2.5 V Bias	Sensors	float	2.5 V	2.45 V	2.55 V			Yes	No
WIB	WIB 1.8 V Bias	Sensors	float	1.8 V	1.75 V	1.85 V			Yes	No
WIB	WIB 1.2 V Current	Sensors	float	10.0 A	8.0 A	12.0 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 3.3 V Current	Sensors	float	9.0 A	7.0 A	11.0 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 0.9 V Current	Sensors	float	2.5 A	1.5 A	3.5 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 0.85 V Current	Sensors	float	13.0 A	11.0 A	15.0 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 5.0 V Current	Sensors	float	1 mA	0 mA	100 mA	Yes	Outside Sepoints	Yes	No
WIB	WIB 2.5 V Current	Sensors	float	5.0 A	3.5 A	6.5 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 1.8 V Current	Sensors	float	4.0 A	3.0 A	5.0 A	Yes	Outside Sepoints	Yes	No
WIB	FEMB1 on/off	FEMBEnable	boolean						Yes	Yes
WIB	FEMB2 on/off	FEMBEnable	boolean						Yes	Yes
WIB	FEMB3 on/off	FEMBEnable	boolean						Yes	Yes
WIB	FEMB4 on/off	FEMBEnable	boolean						Yes	Yes
WIB	Timing Endpoint Status	TimingEndpoint	Uint8						Yes	No
FEMB	FEMB Bias Voltage	Sensors	float	5 V	4.95 V	5.05 V			Yes	No
FEMB	FEMB Bias Current	Sensors	float	1 mA	0 mA	10 mA	Yes	Outside Sepoints	Yes	No
FEMB	Voltage for LArASIC	Sensors / FEMBPower	float	4 V	2.5 V	4.5 V			Yes	Yes
FEMB	Current for LArASIC	Sensors	float	0.4 A	0.3 A	0.9 A	Yes	Outside Sepoints	Yes	No
FEMB	Voltage for ColdADC	Sensors / FEMBPower	float	4 V	2.5 V	4.5 V			Yes	Yes
FEMB	Current for ColdADC	Sensors	float	1.6 A	1.4 A	1.9 A	Yes	Outside Sepoints	Yes	No
FEMB	Voltage for COLDATA	Sensors / FEMBPower	float	4 V	2.5 V	4.5 V			Yes	Yes
FEMB	Current for COLDATA	Sensors	float	0.2 A	0.1 A	0.3 A	Yes	Outside Sepoints	Yes	No

Alarm and Response: PDU example

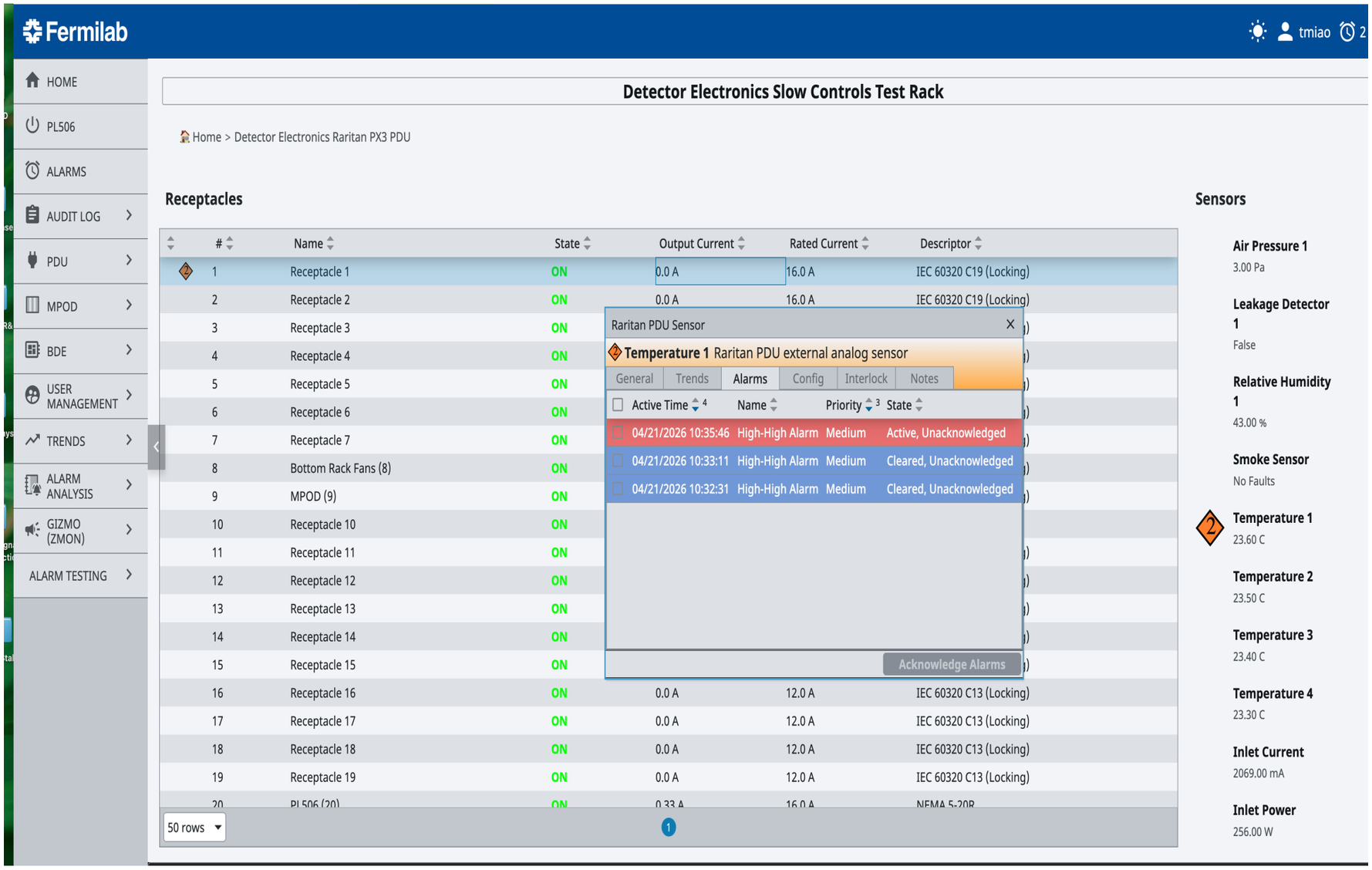




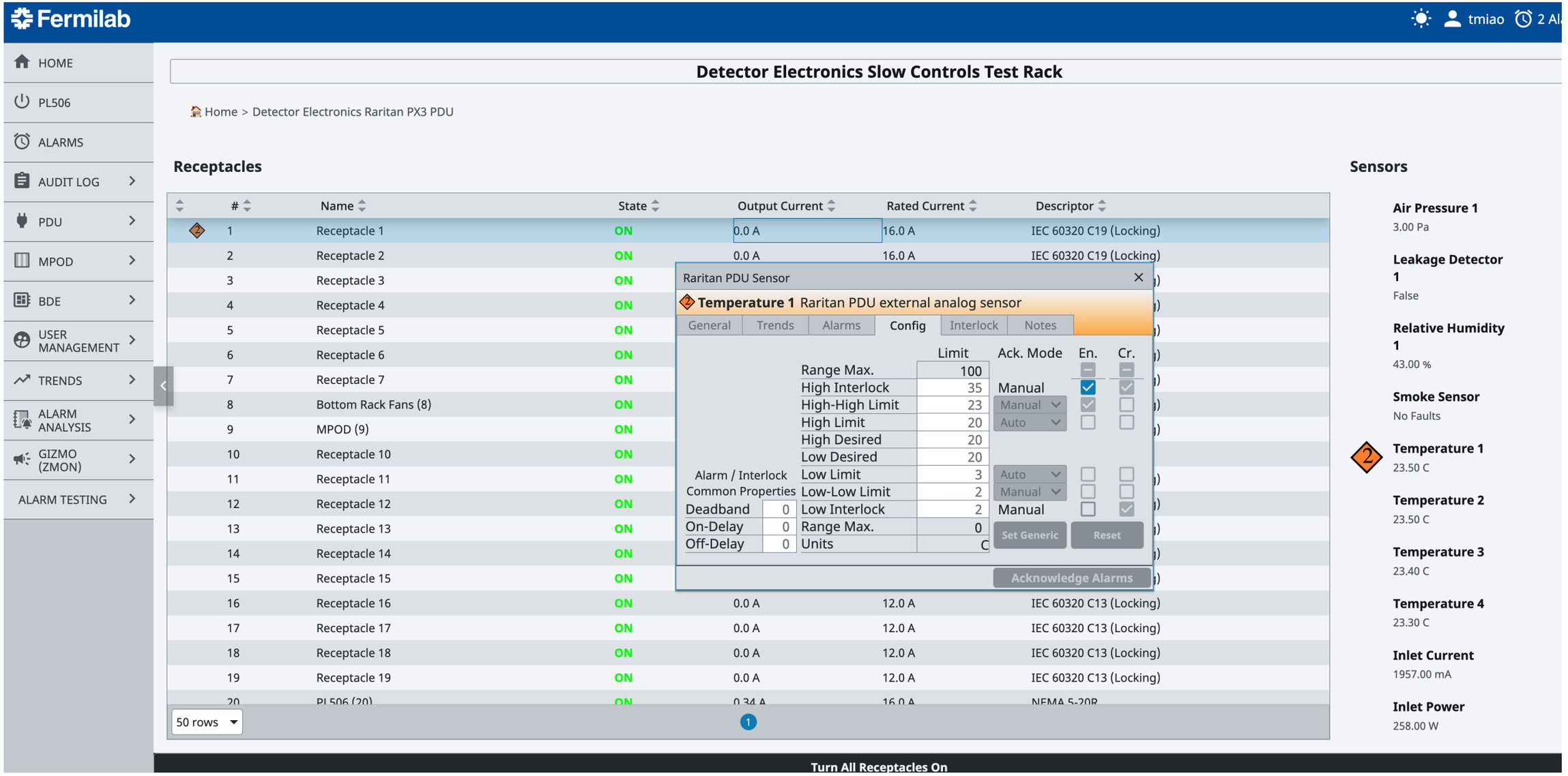
Alarm and Response: temperature sensor reading



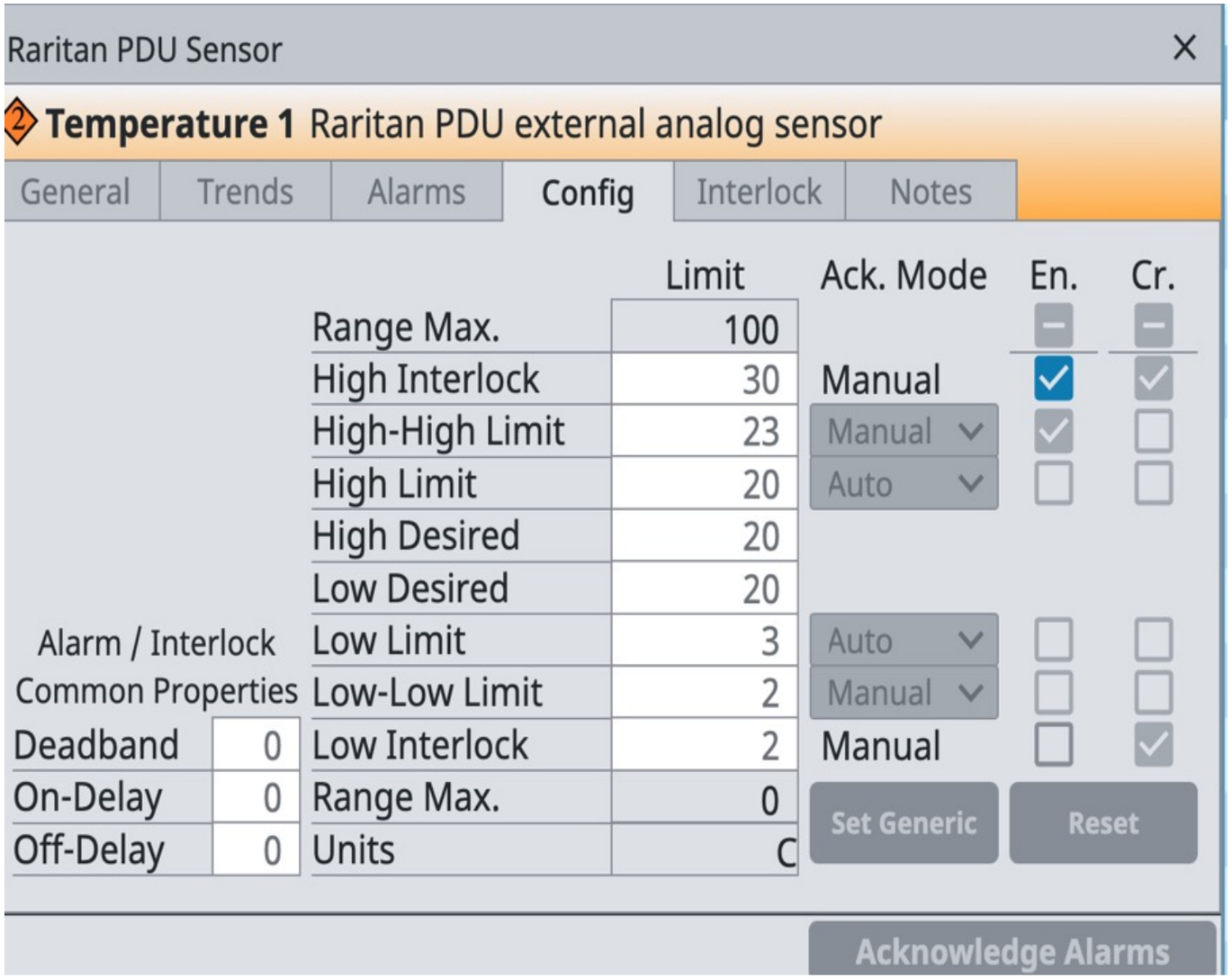
Alarm log: PDU temperature sensor alarm report log



Alarm Response: PDU temperature sensor alarm threshold (re)setting



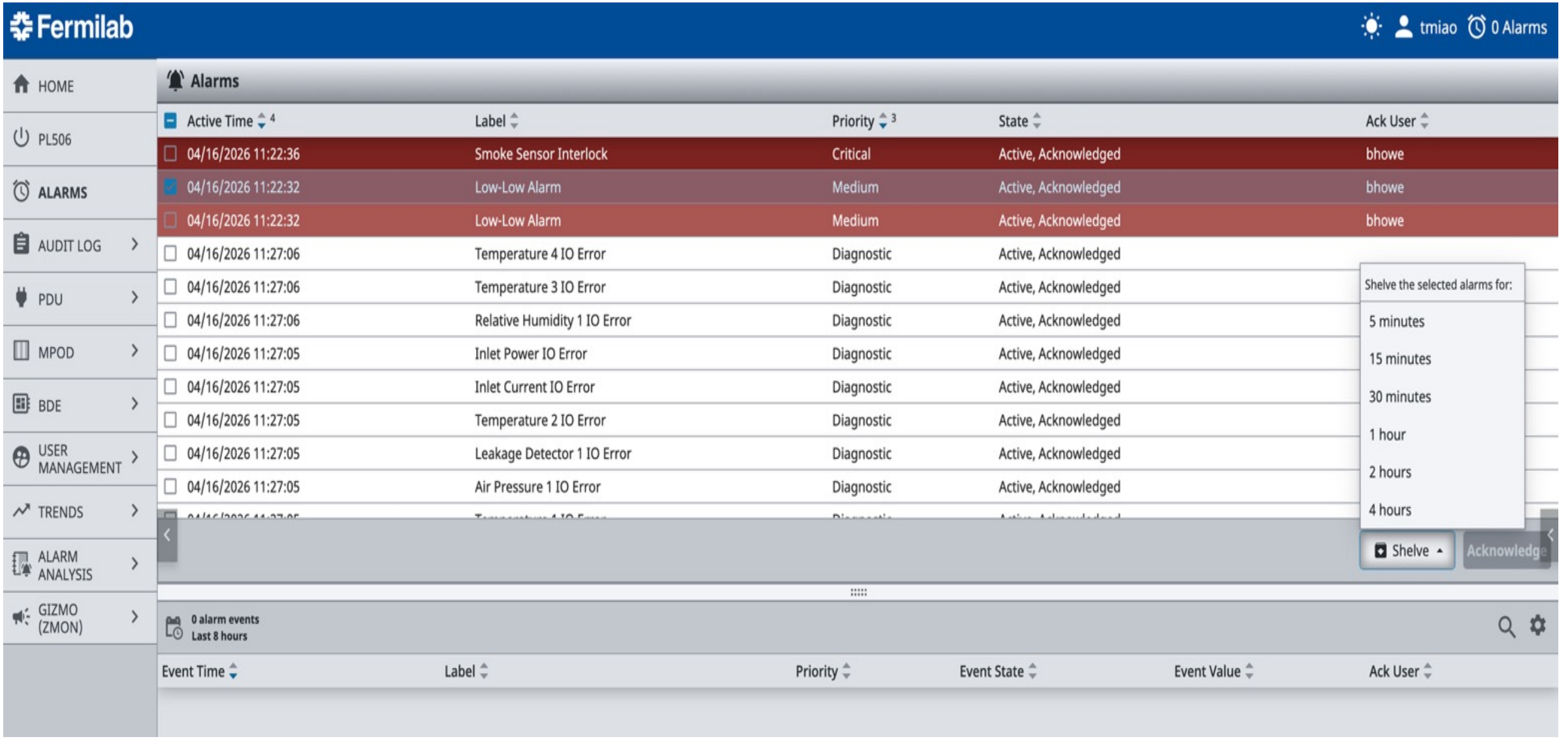
Faceplate for Alarm Response



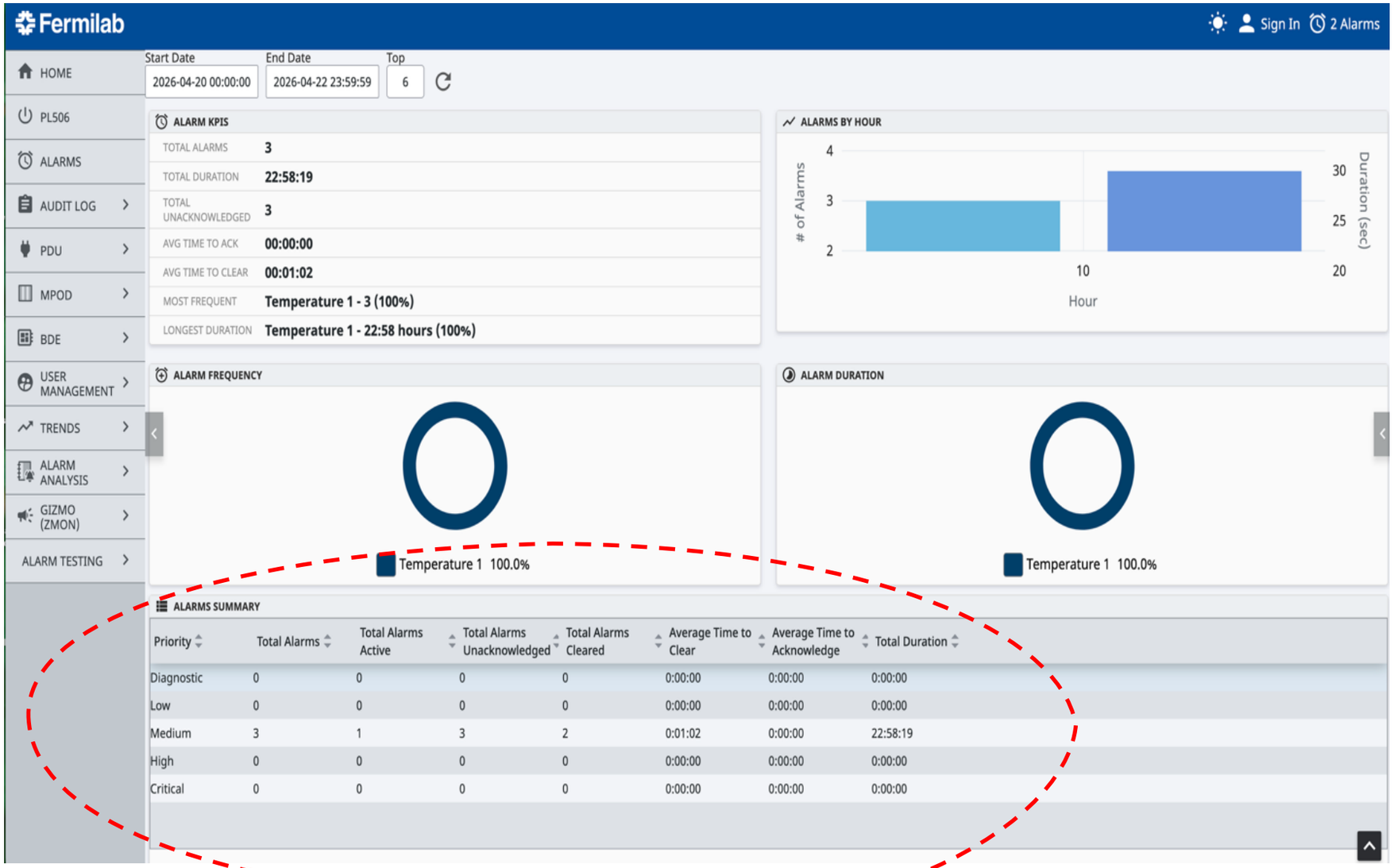
Alarm Threshold Setting:

- High Desired/Low Desired: expected value or ranges
- High Limit / Low Limit: low priority (notification only)
- High-High/Low-Low Limit: medium priority (notification)
- High Interlock / Low Interlock: critical priority alarm threshold (some action required, i.e. PLC interlock to trip/shut down)

Alarm List and Response (shelving)



Alarm Response Status Summary



Need real hardware to test alarm response functions to

- improve existing template
- find balanced approach to handle large number of warnings/alarms from ~1700 devices

Database in Slow Controls

- There are two databases we are planning
  - **Slow controls data** of monitoring parameters by Ignition tag historian
  - **Detector status report data** – summary of detector status including key parameters values (drift HV, bias voltages…)
- Slow controls data storage by tag historian
  - The size of the database is quite large especially at the installation and commissioning stages
  - Tag historian logs tag values based on changes or time intervals – size will be reduced a lot when in operation
- Detector status report data: key monitoring parameters, subsystem alarm and interlock status etc.
  - Accessed by DAQ data quality monitoring (DQM) and offline analysis
  - Will be generated from quick analysis within Ignition: average, min/max of the parameters etc.

Database Servers

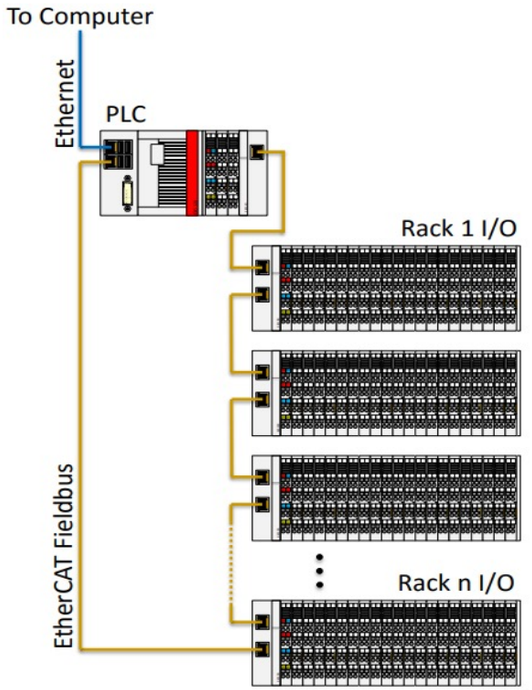
- The hardware for tag historian database servers can be at SURF or Batavia site
  - DAQ condition database servers likely at SURF's Ross-dry basement
  - FDC cryo control database servers at FCC
  - Ignition stores data using *Store and Forward scheme* to ensure data is recorded reliably— extra local disks on gateways
- The detector status report database server needs to be at SURF
  - DQM needs to access with a short turnaround time. Network connection needs to be robust
- Baseline plan is to have slow control database hardware at SURF
- Database needs to be replicated and backed up regularly to offline storage
- Working with COMP, FDC I&I and scientific computing divisions for the specification, procurement and installation

Database Technologies, APIs and Operation

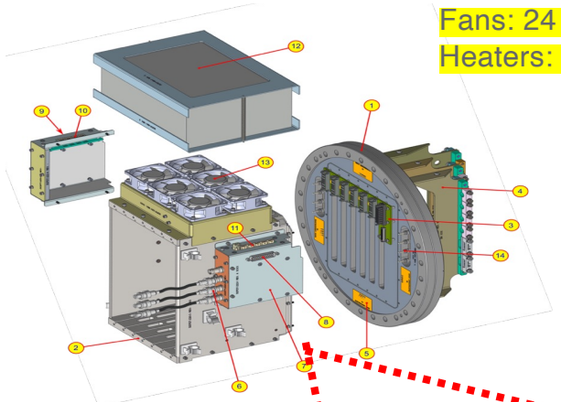
- Ignition works with any relational database
  - Recommendation from Ignition manual is to use 'whichever database your IT department already uses”,
  - that is **PostgreSQL** for Fermilab
- Slow control will also watch DAQ and cryo control’s database technology choice
  - Ignition allows connection to multiple databases
- Counting on database group to provide access APIs: Grafana, Phoebus etc.
  - **Need consortia inputs on their plans and preferred Apps**
- Database operation support from FNAL scientific computing divisions
  - **To also follow SURF overall database operation plan including offline replica scheme**
- We plan to use vertical-slice test stand at IERC to test database operation

Detector Protection PLCs

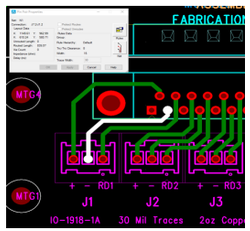
- Detector protection requirements are specified as part of interface documentations
  - Detailed list of interlocks and specifications on PLCs for TPC-e/BDE (also in EDMS 2401090)
  - Good amount specifications from other systems also – **some details still missing**
- We are setting up a DPS test stand using Beckhoff PLC at IERC
  - EtherCAT daisy chain between master PLC and remote I/O
  - PLC master panel with controller to be in one of the 76 electronics racks on side of cryostat
  - Baseline is for remote I/O in the 76 racks (alternative is on cryostat top close to WIECs / DAPHNE…)
- Beckhoff PLC modules for WIEC/PTC/WIB DPS system
  - EL2034 Siemen Sitop PSU8200 24V/40A DCPS for PLCs as well for WIEC fans and flange heaters
  - EL1004 for rotation detector for WIEC fans; EL3202-0010 for 4-wire RDT readout for flange heaters
  - PLC control and logic modules being specified – wants to have one master system for All DPS remote I/Os
  - Setting up a prototype test system at IERC with WIEC fans/heaters

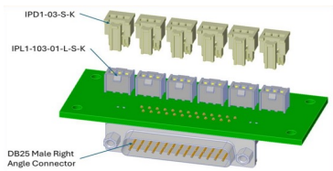


WIEC Fans/Heaters and DPS PLC

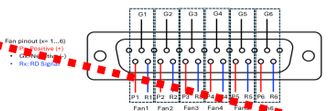


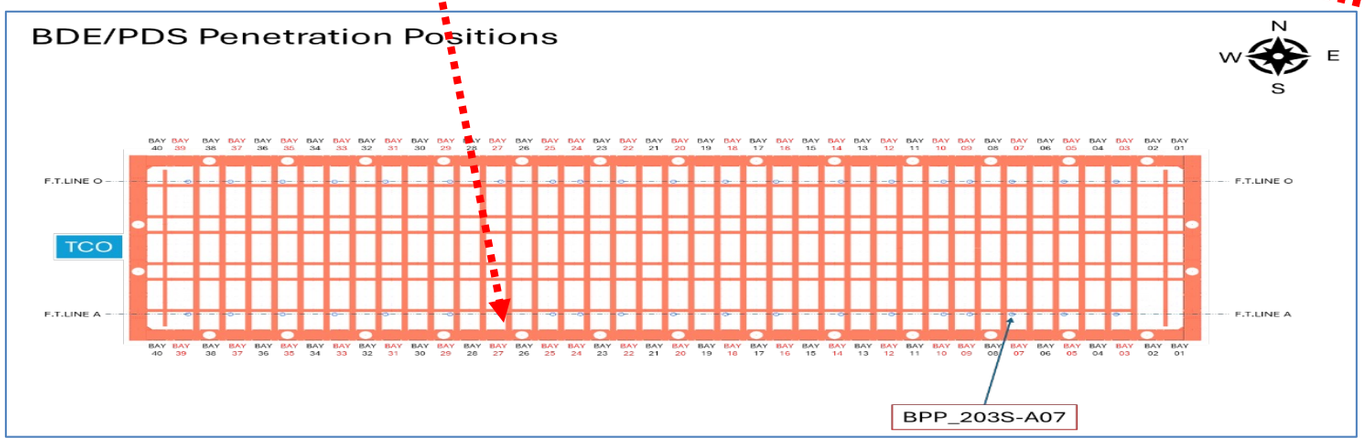
Fans: 24 VDC, 0.33A
Heaters: 24 VDC, 1.42A



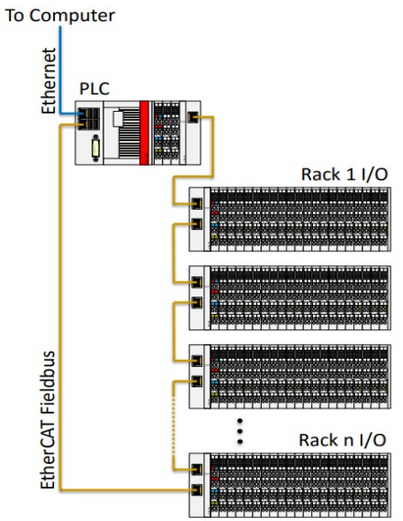


https://edms.cern.ch/document/3342150/2

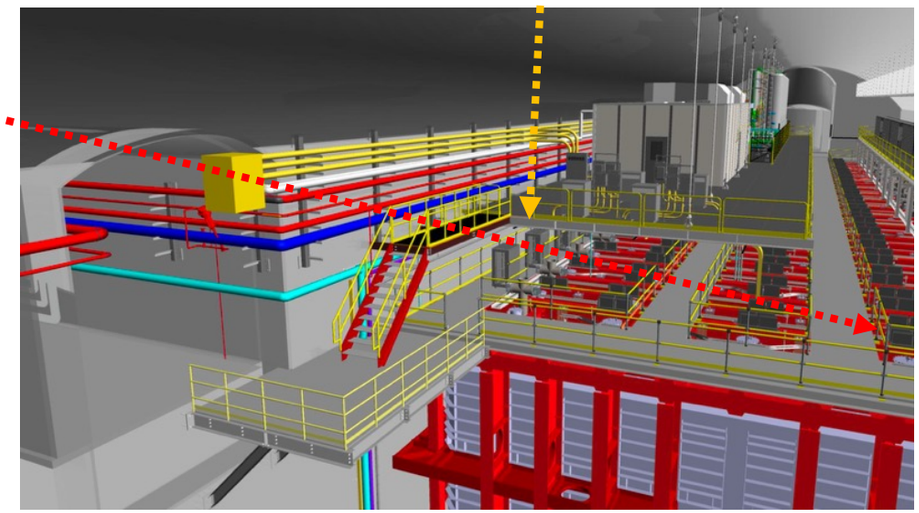




There are 160 cables from PLCs to WIEC's fans/heaters

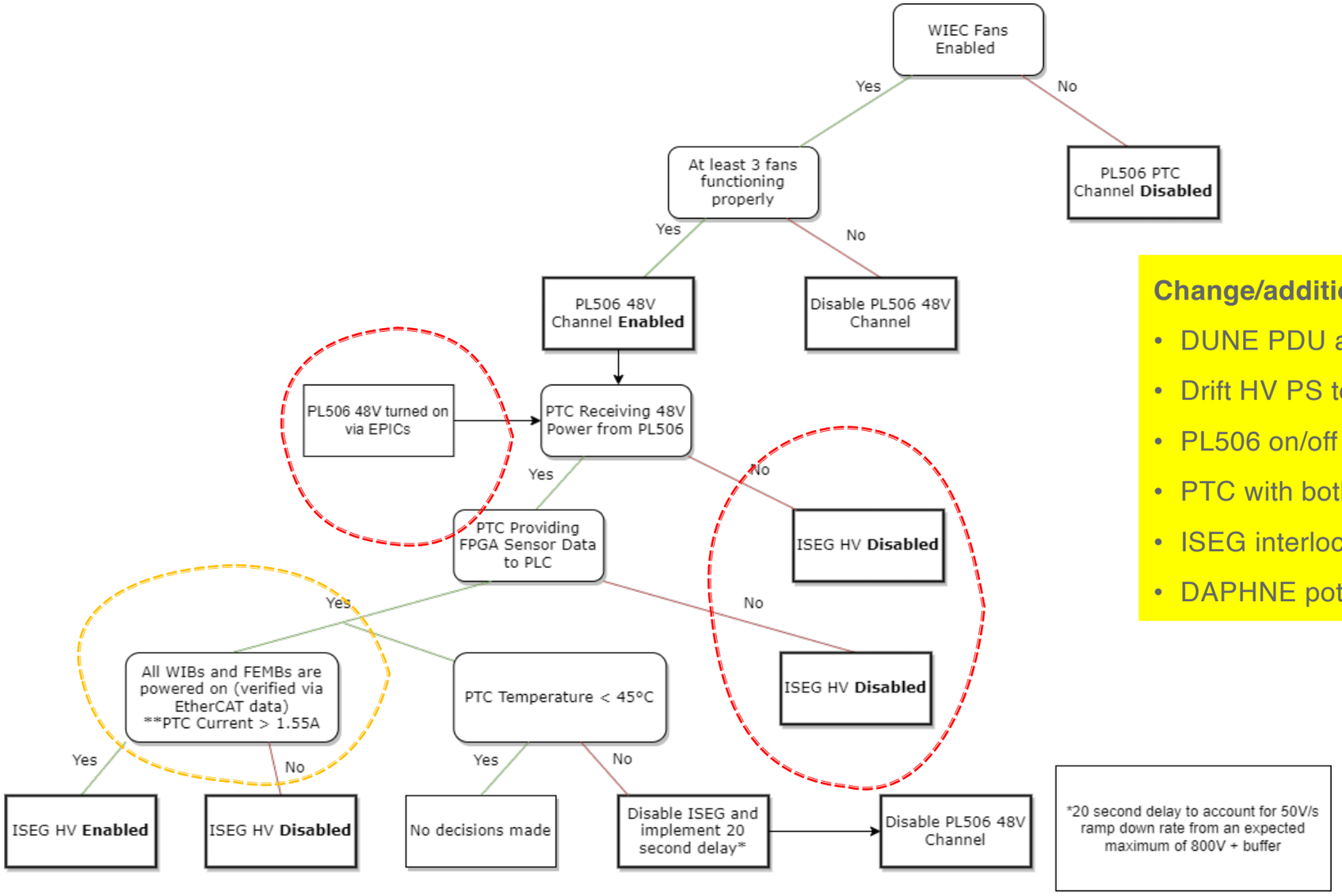


Move PLC I/Os to mini racks near FT?



22AWG DB25 cable with 1.42A has 4.5V voltage drop over 30m

DPS PLC Prototyping ICEBERG→IERC

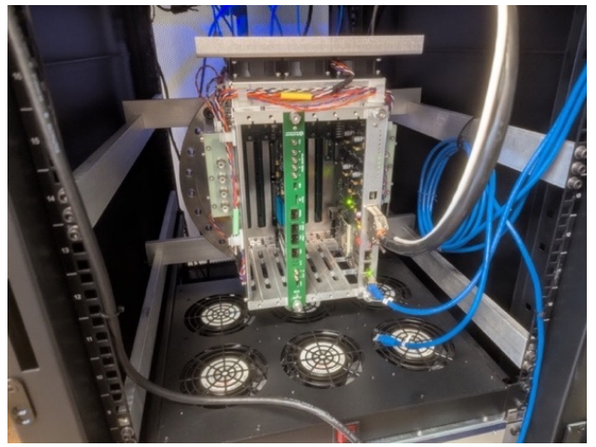


Change/addition from ICEBERG to FD

- DUNE PDU and rack protection sensors
- Drift HV PS to be Heinzinger
- PL506 on/off by DCS Ignition
- PTC with both EtherCat and OPC-UA interfaces
- ISEG interlock (as for FC/CRP bias voltages)
- DAPHNE potentially to be included

Detector Protection System Prototyping at IERC

- WIEC crates with fans and heaters coming to FNAL
  - Production PTC and WIBs couple of months later
- A few ISEG modules available to start CRP/FC bias project
  - ISEG_EHS 8420p/n to be available in September
  - Those ISEG with individual channel interlock ~ cables for 576 channels for CRP/FC
- Heinzinger PS and cRIO will receive interlock signal too
- Need to work out with consortia on cabling production
  - Cables between PLC logic I/O and control devices
- Same test-stand will also be used for SFC automation process and overall architecture layout studies (remote tag, DAQ messaging)





Moving Forward to Final Design and Test

– tasks and interfaces

Design and Test Implementation: device connectivity

- OPC-UA server/client protocols from FD subsystems
  - SNMP→OPC-UA bridge: *Wiener/ISEG PS, PDU*, network switches, GPS-OS
  - ZMQ→OPC-UA bridge: *WIB,* **PTC, DAPHNE**, *GizMon,* **CRP/FC level meters**
  - **IPMI→OPC-UA** bridge: DAQ/SC servers, uTCA/MCH for TDE and DAQ timing
  - OPC-UA on the board: DAQ GPS-Interface-Board (GIB)
  - PLC: detector protection, **PoF laser box, cold cameras and LEDs**, LAr level transmitter
  - **NI LabVIEW →OPC-UA bridge**: Heinzinger/cRIO, RTD/cRIO, PrM/Digitizer
  - **BACnet** of Metasys Building Automation and Control Network
  - Some not yet specified: **FBG interrogator** I4-16 from optics'11, temperature sensors inside cryostat insulation…
- Important to test all OPC-UA implementation to ensure they work correctly
  - The starting point is to have board-level control code (i.e. ZMQ) available
  - ***Slow controls has a dedicated team lead by Tom Junk (senior scientist) on OPC-UA***

Device Connectivity and GUI Implementation Status

- BDE implementation started with electronics at PAB
  - Also setting a vertical test-stand with WIEC/WIB/PTC/PL506 at IERC
  - The test-stand is critical to test all aspects of DCS design: monitoring, alarm, SFC, database
- New project also getting started
  - CRP/FC bias voltage project using couple of ISEG modules
  - IPMI→OPC-UA with the 2 rack-mount servers with BMC: using IPMItool for DAQ/SC servers, uTCA crates of DAQ timing and TDE
- Expect to start this summer
  - A 150KV Heinzinger PS is at FNAL; cRIO electronics arriving in June/July
  - DAQ timing control devices is scheduled to arrive Fermilab late this summer
  - Working with PDS to identify control devices for PoF laser boxes
  - DAPHNE waits for ZMQ implementation – new consortium people to help
- Need consortium to provide people to work on connectivity and GUI
  - as resource plan outlined by Eric J in 2-11-2026 TB meeting

Hardware and Software Delivery for Prototyping Test

- BDE: WIB/PTC/PL506 in ICEBERG, PL506 at IERC
  - WIEC with fans and heaters to ship to FNAL soon; DB25 cabling between fans/heaters and PLC?
  - PTC ZMQ code by Sihon Kubota is close to be ready; OPC-UA server code will be done by Tom Junk
  - Consortium to send a PTC and a few WIBs for vertical slice test-stand to FNAL in August/Sept
  - Need consortium expert to participate alarm response, SFC process, detector status report database
- HVS: 150KV Heinzinger PS is at hand, cRIO electronics arriving in June/July
  - Need DB15 cable from Heinzinger analogue control → cRIO: Xavier is making them?
  - LabVIEW VIs from CERN, need help on NI LabVIEW OPC-UA kit and Ignition integration test
  - Adding more people on CRP/FC bias GUI and SFC implementation
  - Need design/review for cold camera/LED controller (heater): Simen S7-1500 → Beckhoff PLC?
  - Need people on GUI development for Heinzinger, cold camera/LED (FSM in LabVIEW)
  - Need long term LabVIEW support for installation and operation for FDC – looking Fermilab support

Hardware and Software Delivery for Prototyping Test at IERC

- PDS: MPV, DAPHNE at Fermilab, PoF laser control to be reconfigured for test at IERC
  - Work to choose PoF laser control device: Arduino Opta PLC vs. Beckhoff PLC (Opta delivery late, keep pushing)
  - New people to start ZMQ code on DAPHNE to start OPC-UA (board is available at FNAL)
  - Need consortium to assign people on GUI work (template from BDE a good start point)
  - There is still question about the need to monitor/control siPM power line from DAPHNE
- TDE
  - Need uTCA crate with MCH
  - WIENER MPV available to start power distribution box control GUI
  - Calibration box: Raspberry Pi → OPC-UA?
  - No documentation on detector protection requirement yet

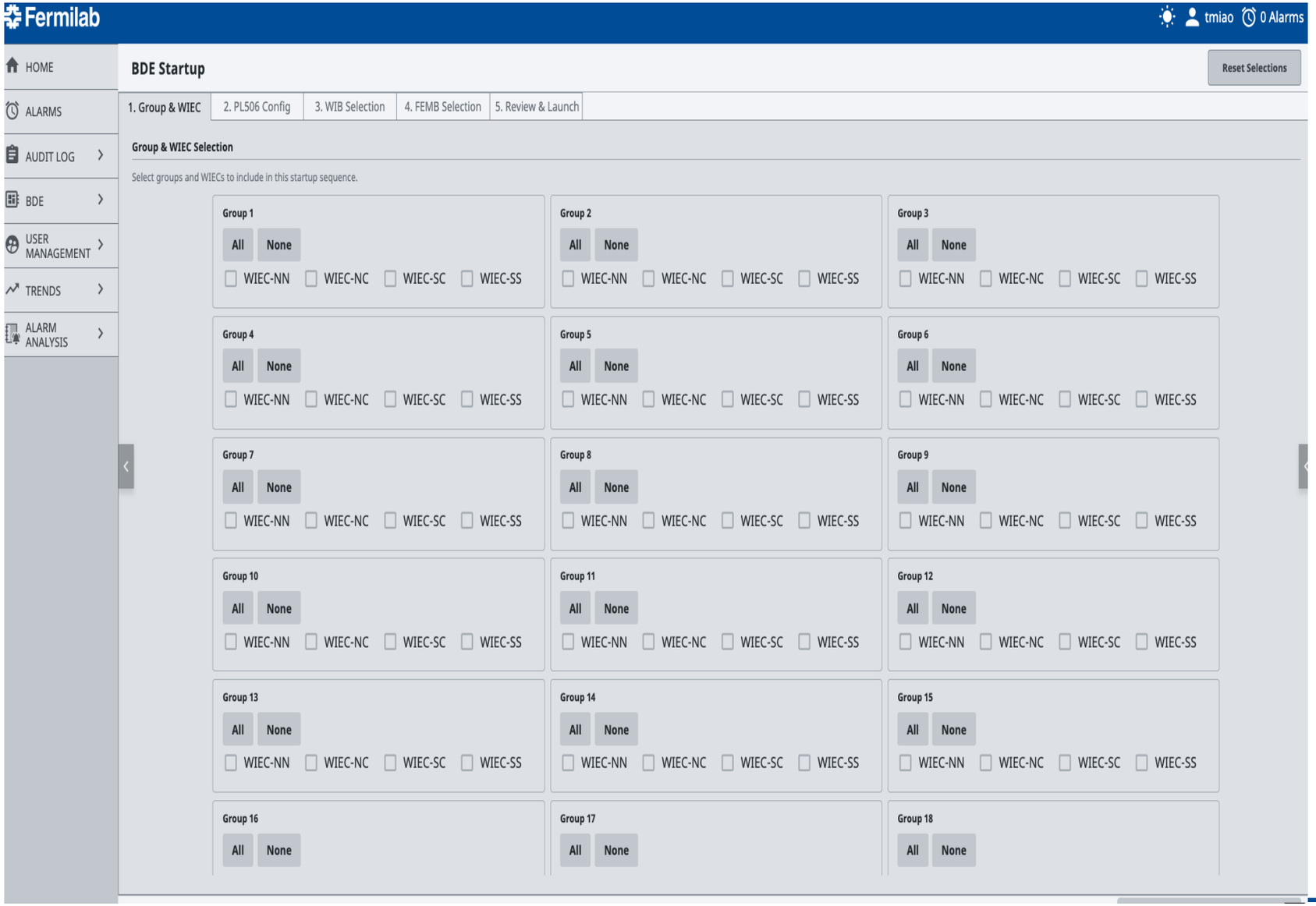
Hardware and Software Delivery for Prototyping Test at IERC

- DAQ:
  - OPC-UA on GIB ready, waiting GIB to start GUI using Ignition – GIB delivery to Fermilab September?
  - SNMP→OPCUA using Kymera bridge. Waiting for GPS-OS – deliver September?
  - IPMItool →OPC-UA on DAQ servers: to start with newly delivered SuperMicro servers at IERC G271
  - need a Cisco network switch to start GUI on readout network switch monitoring (SNMP→OPC-UA, MRTG traffic monitoring)
  - need to assign DAQ consortium people to work on GUI for monitoring work
  - will start DAQ-SC messaging test with WIB/PTC/PLC vertical slice test stand this fall
- CaCI
  - need to finish the interface documentation and make plans for prototyping efforts
  - Likely will need AlarzTech digitizer/NI DAQ, LabVIEW VI for PrM and RTD, FBG interrogator
  - Iolaser control has an OPC-UA using Quasar framework (Nuno Barros)

System Level Design and Test at Fermilab

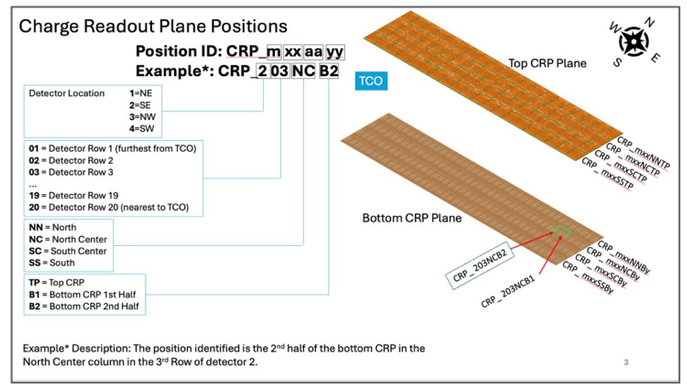
- Automation process using SFC – to operate large size subsystems
- Ignition gateway scale-out layout architecture
  - “remote” tag creation and sharing between back-end and front-end gateways
  - Tag historian database: redundant copy on local disk, offline replica, API access, offline analysis
- Slow controls – DAQ messaging
  - Baseline protocol for messaging is OPC-UA server/client to pass along slow control status and DAQ CCM information
  - Other messaging protocol is also to be explored
  - DAQ plans to run prototype DAQ applications on IERC Linux box
- Slow control status data to DQM
  - Likely a separate database in addition to tag historian database
  - Will work with COMP and FNAL scientific computing to use database connection available on-site at Batavia

SFC BDE Startup Page: select of WIECs to power on



Four WIECs for a row of CRPs – mapping by ManHong Z

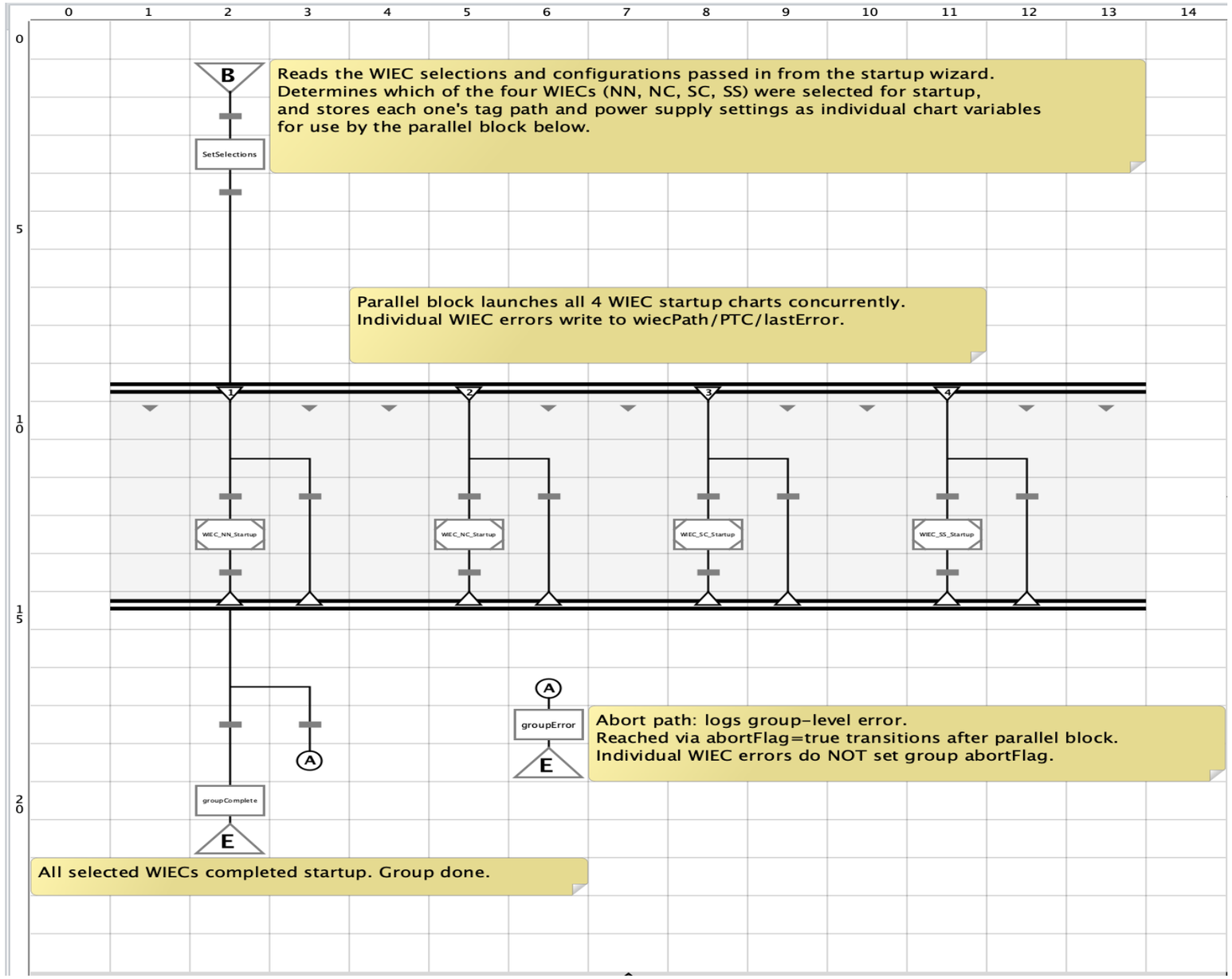
WIEC-01NN	WIEC-01NC	WIEC-01SC	WIEC-01SS	PL506-01	Power on/off CRP Row 1	DPS PLC I/O-01
WIEC-02NN	WIEC-02NC	WIEC-02SC	WIEC-02SS	PL506-02	Power on/off CRP Row 2	
WIEC-03NN	WIEC-03NC	WIEC-03SC	WIEC-03SS	PL506-03	Power on/off CRP Row 3	DPS PLC I/O-02
WIEC-04NN	WIEC-04NC	WIEC-04SC	WIEC-04SS	PL506-04	Power on/off CRP Row 4	
WIEC-05NN	WIEC-05NC	WIEC-05SC	WIEC-05SS	PL506-05	Power on/off CRP Row 5	DPS PLC I/O-03
WIEC-06NN	WIEC-06NC	WIEC-06SC	WIEC-06SS	PL506-06	Power on/off CRP Row 6	
WIEC-07NN	WIEC-07NC	WIEC-07SC	WIEC-07SS	PL506-07	Power on/off CRP Row 7	DPS PLC I/O-04
WIEC-08NN	WIEC-08NC	WIEC-08SC	WIEC-08SS	PL506-08	Power on/off CRP Row 8	
WIEC-09NN	WIEC-09NC	WIEC-09SC	WIEC-09SS	PL506-09	Power on/off CRP Row 9	DPS PLC I/O-05
WIEC-10NN	WIEC-10NC	WIEC-10SC	WIEC-10SS	PL506-10	Power on/off CRP Row 10	



BDE startup SFC is being designed and tested by Brandon H and Trevor N

SFC BDE Startup:

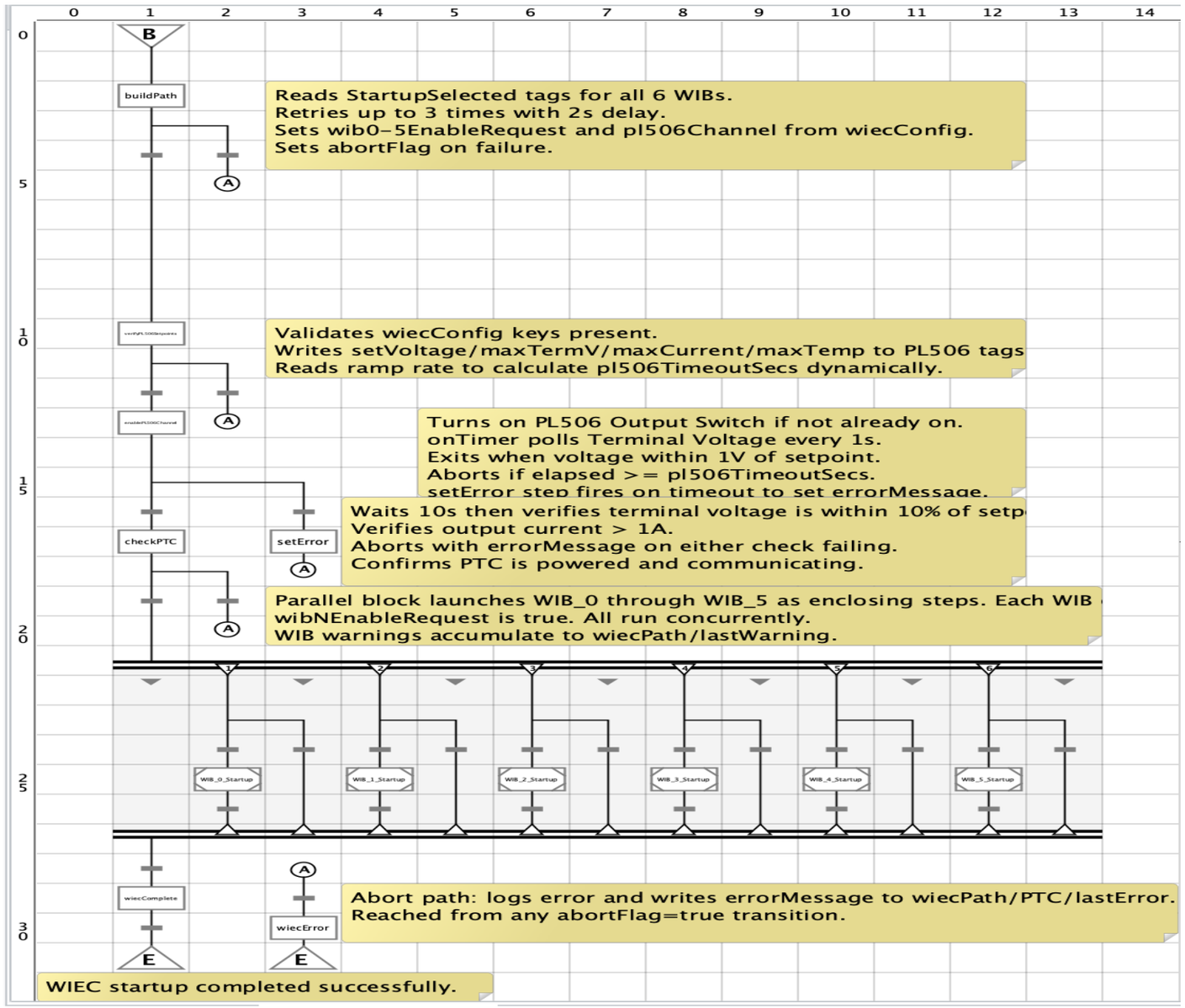
4 WIECs in parallel



SFC BDE Startup:

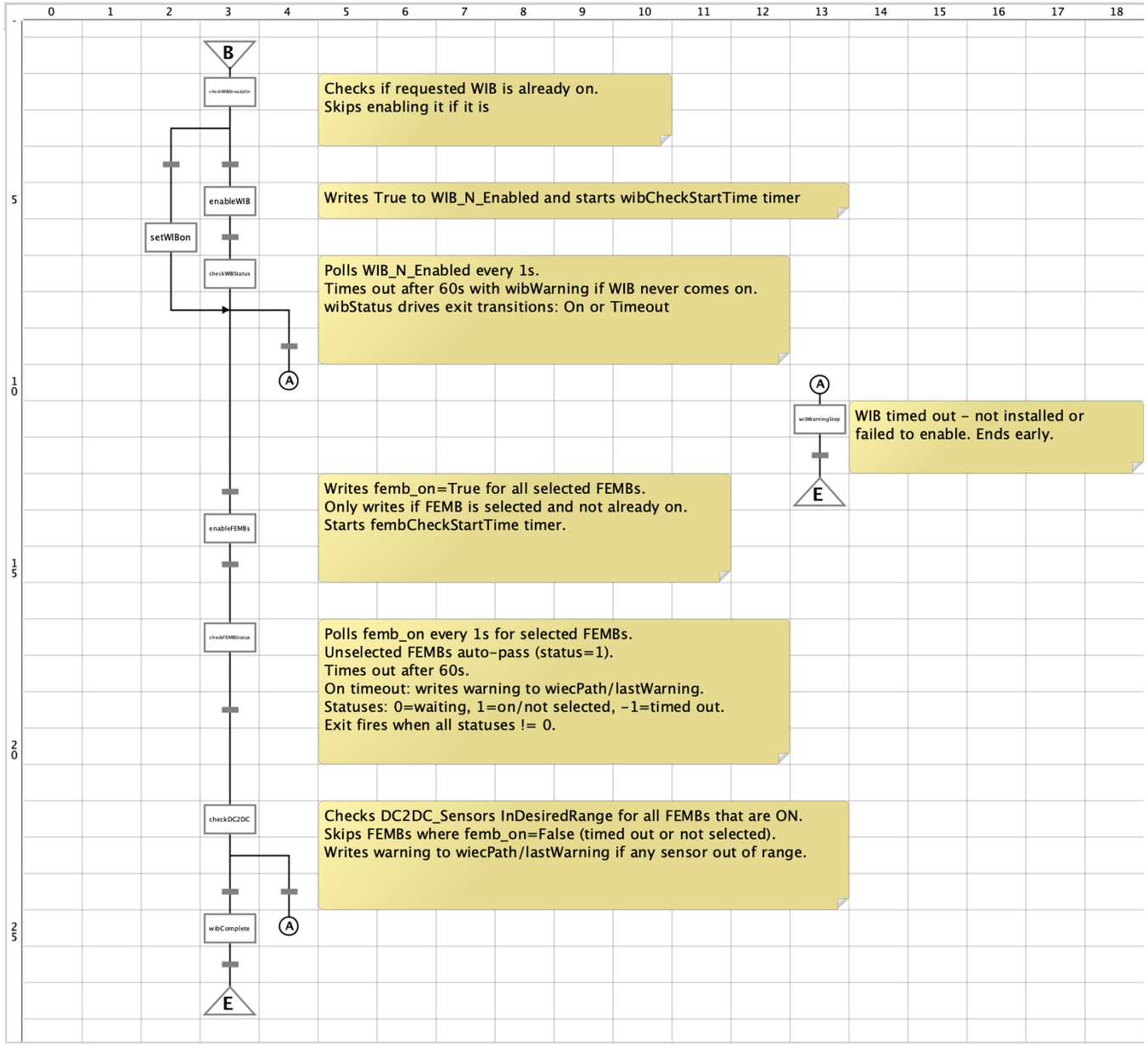
6 WIBs in parallel

(Turn on PL506 first, check temperatures, voltage lines…)



SFC BDE Startup

checking and turning on WIB

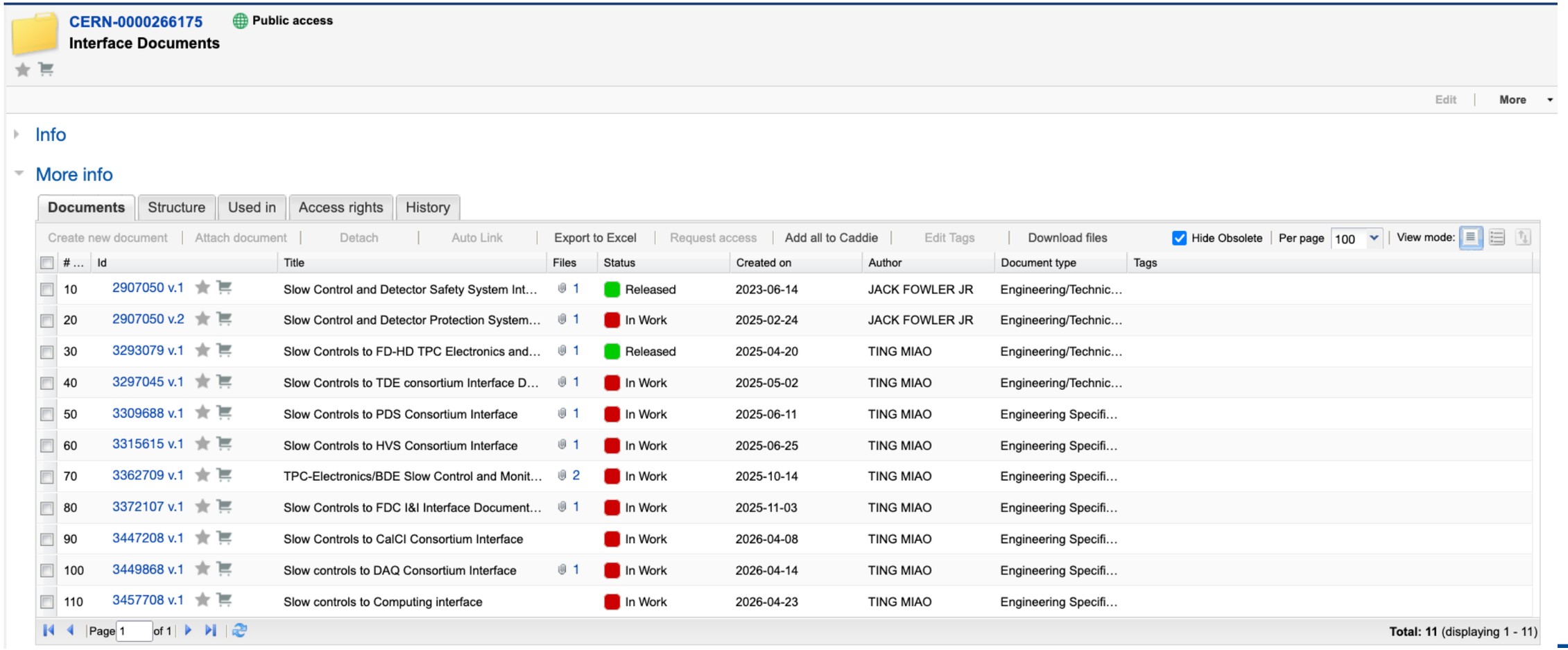


Moving to Final Installation Design

- Coordination with consortia is critical: **devices to FNAL for prototyping test**
  - need to complete test implementation of SFC, data achieving and communication functions to DAQ
  - a vertical slice setup of WIEC/WIB/PTC/PL506/DPS PLC is the most efficient way to move the project along
- Identify people from consortium to work with slow controls group
  - to make sure the OPC-UA servers of your devices correctly expose the data to Ignition
  - to get the basic control and monitoring functions of your device implemented the way useful for test and operation
  - to understand how those functions were implemented, and you can modify when you need to do so
- Identify SURF team for DCS/DPS tasks both on hardware and software
  - need to have an Ignition gateway soon to communicate to building equipment and environmental sensors
  - need to start to monitor/control racks, servers, networks, DAQ devices, grounding as early as summer 2027

Interface Documentation

- ICD written with all consortia and going through expert and management reviews
- Finishing CalCI and COMP before focusing on I&I interfaces
- The documentations are in EDMS: https://edms.cern.ch/project/CERN-0000266175



CERN-0000266175 Public access

Interface Documents

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Info

More info

Documents | Structure | Used in | Access rights | History

Create new document | Attach document | Detach | Auto Link | Export to Excel | Request access | Add all to Caddie | Edit Tags | Download files | Hide Obsolete | Per page 100 | View mode:

# ...	Id	Title	Files	Status	Created on	Author	Document type	Tags
10	2907050 v.1	Slow Control and Detector Safety System Int...	1	Released	2023-06-14	JACK FOWLER JR	Engineering/Technic...	
20	2907050 v.2	Slow Control and Detector Protection System...	1	In Work	2025-02-24	JACK FOWLER JR	Engineering/Technic...	
30	3293079 v.1	Slow Controls to FD-HD TPC Electronics and...	1	Released	2025-04-20	TING MIAO	Engineering/Technic...	
40	3297045 v.1	Slow Controls to TDE consortium Interface D...	1	In Work	2025-05-02	TING MIAO	Engineering/Technic...	
50	3309688 v.1	Slow Controls to PDS Consortium Interface	1	In Work	2025-06-11	TING MIAO	Engineering Specifi...	
60	3315615 v.1	Slow Controls to HVS Consortium Interface	1	In Work	2025-06-25	TING MIAO	Engineering Specifi...	
70	3362709 v.1	TPC-Electronics/BDE Slow Control and Monit...	2	In Work	2025-10-14	TING MIAO	Engineering Specifi...	
80	3372107 v.1	Slow Controls to FDC I&I Interface Document...	1	In Work	2025-11-03	TING MIAO	Engineering Specifi...	
90	3447208 v.1	Slow Controls to CalCI Consortium Interface		In Work	2026-04-08	TING MIAO	Engineering Specifi...	
100	3449868 v.1	Slow controls to DAQ Consortium Interface	1	In Work	2026-04-14	TING MIAO	Engineering Specifi...	
110	3457708 v.1	Slow controls to Computing interface		In Work	2026-04-23	TING MIAO	Engineering Specifi...	

Page 1 of 1

Total: 11 (displaying 1 - 11)

Interface Documentation Status

Consortium	EDMS posted	Reviewed	Approved	Hardware available to start DCS work	Software ready to start DCS work	Note
TPC_e/BDE	3362709 v.1	yes	yes	partially	WIB-yes, PTC- close	WIEC to FNAL in June PTC/WIB in August
HVS	3315615 v.1	yes	no	partially	yes	cRIO delivery in June ISEG in Sept
PDS	3309688 v.1	yes	no	partially	DAPHNE-no, PoF-no	ZMQ for DAPHNE, u)PLC for PoF
TDE	3297045 v.1	yes	no	no	uTCA-yes, Cali-no	IPMItools-->OPCUA
DAQ	3449868 v.2	on-going	no	no	yes	GIB & GPS-OS coming in Sept
CalCI	draft					
COMP	draft					

- We need to review and sign off the interface documentation this summer
  - Follow-up the agreement and provide hardware/software for DCS prototyping
  - Identify detector consortium institutes to work on slow controls: connectivity, GUI…
- With progress of prototyping, we need to look at monitoring parameter list again and provide more details – WIB/PTC parameters in next slides and others in backup slides

WIB Control and Monitoring Parameters

Warm Interface Board and Frontend MotherBoard Control and Monitoring Parameters

Hardware Device	Description	OPC-UA Address Name	Data Type	Nominal Value	Lower Bound	High Bound	Included in alarm table	Alarm Mode	Can be Monitored (Yes/No)	Configurable by DCS (Yes/No)
WIB	Firmware version	GetFWVersion	Uint32						Yes	No
WIB	WIB Status Bits	GetWIBStatus	Uint32						Yes	No
WIB	Temperature Sensors	Sensors	float	60 C	30 C	80 C	Yes	Above Setpoint	Yes	No
WIB	WIB 1.2 V Bias	Sensors	float	1.2 V	1.15 V	1.25 V			Yes	No
WIB	WIB 3.3 V Bias	Sensors	float	3.3 V	3.25 V	3.35 V			Yes	No
WIB	WIB 0.9 V Bias	Sensors	float	0.9 V	0.85 V	0.95 V			Yes	No
WIB	WIB VCC_PSPLL 1.2 V Bias	Sensors	float	1.2 V	1.15 V	1.25 V			Yes	No
WIB	WIB DDR4_VTT 0.6 V Bias	Sensors	float	0.6 V	0.55 V	0.65 V			Yes	No
WIB	WIB 0.85 V Bias	Sensors	float	0.85 V	0.80 V	0.90 V			Yes	No
WIB	WIB 5.0 V Bias	Sensors	float	5 V	4.95 V	5.05 V			Yes	No
WIB	WIB 2.5 V Bias	Sensors	float	2.5 V	2.45 V	2.55 V			Yes	No
WIB	WIB 1.8 V Bias	Sensors	float	1.8 V	1.75 V	1.85 V			Yes	No
WIB	WIB 1.2 V Current	Sensors	float	10.0 A	8.0 A	12.0 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 3.3 V Current	Sensors	float	9.0 A	7.0 A	11.0 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 0.9 V Current	Sensors	float	2.5 A	1.5 A	3.5 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 0.85 V Current	Sensors	float	13.0 A	11.0 A	15.0 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 5.0 V Current	Sensors	float	1 mA	0 mA	100 mA	Yes	Outside Sepoints	Yes	No
WIB	WIB 2.5 V Current	Sensors	float	5.0 A	3.5 A	6.5 A	Yes	Outside Sepoints	Yes	No
WIB	WIB 1.8 V Current	Sensors	float	4.0 A	3.0 A	5.0 A	Yes	Outside Sepoints	Yes	No
WIB	FEMB1 on/off	FEMBEnable	boolean						Yes	Yes
WIB	FEMB2 on/off	FEMBEnable	boolean						Yes	Yes
WIB	FEMB3 on/off	FEMBEnable	boolean						Yes	Yes
WIB	FEMB4 on/off	FEMBEnable	boolean						Yes	Yes
WIB	Timing Endpoint Status	TimingEndpoint	Uint8						Yes	No
FEMB	FEMB Bias Voltage	Sensors	float	5 V	4.95 V	5.05 V			Yes	No
FEMB	FEMB Bias Current	Sensors	float	1 mA	0 mA	10 mA	Yes	Outside Sepoints	Yes	No
FEMB	Voltage for LArASIC	Sensors / FEMBPower	float	4 V	2.5 V	4.5 V			Yes	Yes
FEMB	Current for LArASIC	Sensors	float	0.4 A	0.3 A	0.9 A	Yes	Outside Sepoints	Yes	No
FEMB	Voltage for ColdADC	Sensors / FEMBPower	float	4 V	2.5 V	4.5 V			Yes	Yes
FEMB	Current for ColdADC	Sensors	float	1.6 A	1.4 A	1.9 A	Yes	Outside Sepoints	Yes	No
FEMB	Voltage for COLDATA	Sensors / FEMBPower	float	4 V	2.5 V	4.5 V			Yes	Yes
FEMB	Current for COLDATA	Sensors	float	0.2 A	0.1 A	0.3 A	Yes	Outside Sepoints	Yes	No

PTC Control and Monitoring Parameters

Hardware Device	Description	OPC-UA Address Name	Data Type	Nominal Value	Lower Bound	High Bound	Included in alarm table	Alarm Mode	Can be Monitored (Yes/No)	Configurable by DCS (Yes/No)
PTC	Firmware version	GetPTCFWVersion	Uint32						Yes	No
PTC	Timing Status	TimingEndpoint	Uint32						Yes	No
PTC	Temperature Sensors	Sensors	float				yes	Above Setpoint	Yes	No
PTC	WIB 1 Enabled	WIBPower	boolean						Yes	Yes
PTC	WIB 2 Enabled	WIBPower	boolean						Yes	Yes
PTC	WIB 3 Enabled	WIBPower	boolean						Yes	Yes
PTC	WIB 4 Enabled	WIBPower	boolean						Yes	Yes
PTC	WIB 5 Enabled	WIBPower	boolean						Yes	Yes
PTC	WIB 6 Enabled	WIBPower	boolean						Yes	Yes
PTC	WIB 1 Voltage	Sensors	float	12V			yes	Outside Setpoints	Yes	No
PTC	WIB 1 Current	Sensors	float				yes	Outside Setpoints	Yes	No
PTC	WIB 2 Voltage	Sensors	float	12V			yes	Outside Setpoints	Yes	No
PTC	WIB 2 Current	Sensors	float				yes	Outside Setpoints	Yes	No
PTC	WIB 3 Voltage	Sensors	float	12V			yes	Outside Setpoints	Yes	No
PTC	WIB 3 Current	Sensors	float				yes	Outside Setpoints	Yes	No
PTC	WIB 4 Voltage	Sensors	float	12V			yes	Outside Setpoints	Yes	No
PTC	WIB 4 Current	Sensors	float				yes	Outside Setpoints	Yes	No
PTC	WIB 5 Voltage	Sensors	float	12V			yes	Outside Setpoints	Yes	No
PTC	WIB 5 Current	Sensors	float				yes	Outside Setpoints	Yes	No
PTC	WIB 6 Voltage	Sensors	float	12V			yes	Outside Setpoints	Yes	No
PTC	WIB 6 Current	Sensors	float				yes	Outside Setpoints	Yes	No

PL506 Control and Monitoring Parameters

Hardware Device	OPC-UA Address Name	Data Type	Nominal Value	Lower Bound	High Bound	Monitoring Only (Yes/No)	Configurable by DCS (Yes/No)	Included in alarm table	Alarm Mode
PL506	Serial number	string							
PL506	Total operating time	Uint32							
PL506	Measured sense voltage	float				Yes	No		
PL506	Measured terminal voltage	float				Yes	No		
PL506	Measured output current	float				Yes	No	yes	Outside Setpoints
PL506	Measured temperature	float						yes	Above Setpoint
PL506	Sense voltage set point	float	48V			Yes	Yes		Outside Setpoints
PL506	Current Limit	float	10A			Yes	Yes		
PL506	Voltage ramp up rate	float	1 V/S			Yes	Yes		
PL506	Voltage ramp down rate	float				Yes	Yes		
PL506	Maximum temperature	float							
PL506	Output status	boolean				Yes	No		
	temperature sensors readings								
temperature warning/failure threshold setting									
	Fan Speed							yes	Outside Setpoints
	Fan Speed RPM setting							yes	Outside Setpoints
	Fan Air Temperature							yes	Above Setpoint

Summary

- Good progress on design and test since last FDC I&I meeting
- Technical progress is steady; working to get more devices for prototyping
- Final design and test phase to start this summer at Fermilab
- Critical to continue work closely with consortia to plan DCS/DPS activities
- Need more people getting involved